Monotonicity

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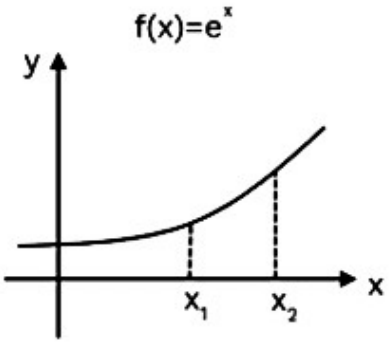
Monotonicity

GENERAL INTRODUCTION :

The most useful element taken into consideration amongst the total postmortem activities of functions, is their monotonic behaviour.

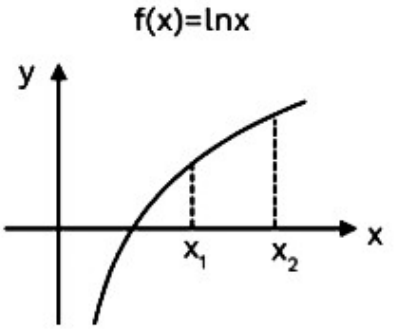
(a) Monotonic function :

Functions are said to be monotonic if they are either increasing or decreasing in their entire domain e.g. $f(x)=e^x$; $f(x)=\ln x$ & $f(x)=2x+3$ are some of the examples of functions which are increasing whereas $f(x)=-x^3$; $f(x)=e^{-x}$ and $f(x)=\cot^{-1}(x)$ are some of the examples of the functions which are decreasing.



Increasing function

If $x_1 < x_2 \Leftrightarrow f(x_1) < f(x_2)$ then function is called increasing function or strictly increasing function.

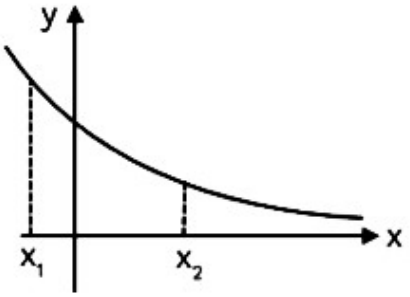


Decreasing function

$f(x) = e^{-x}$

If $x_1 < x_2$

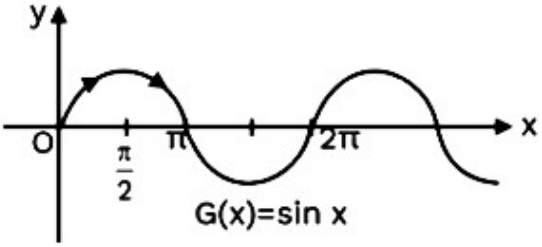
but $f(x_1) > f(x_2)$ in entire domain then function is called decreasing function or strictly decreasing function.

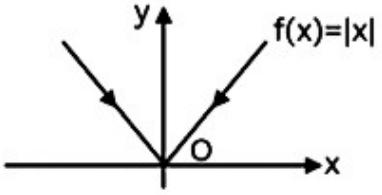


(b) Non-Monotonic :

Functions which are increasing as well as decreasing in their domain are said to be non-monotonic e.g. $f(x) = \sin x$; $f(x) = ax^2 + bx + c$ and $f(x) = |x|$, however in the interval $\left[0, \frac{\pi}{2}\right]$,

$f(x)=\sin x$ will be said to be increasing.



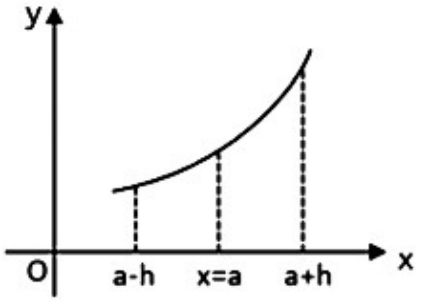


Monotonicity of a function at a point :

A function is said to be monotonically increasing at x = a

if f(x) satisfies

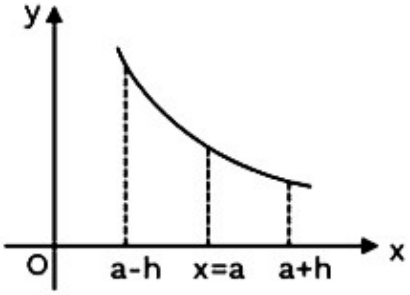
$$\left.\begin{matrix} f(a+h) > f(a) \\ f(a-h) < f(a) \end{matrix}\right] \text{ for a small Positive h}$$



A function is said to be monotonically decreasing at

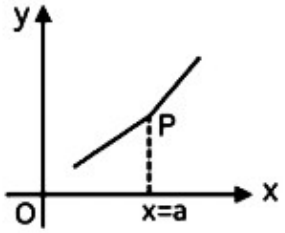
x = a if f(x) satisfies

$$\left.\begin{matrix} f(a+h) < f(a) \\ f(a-h) > f(a) \end{matrix}\right] \text{ for a small Positive h.}$$



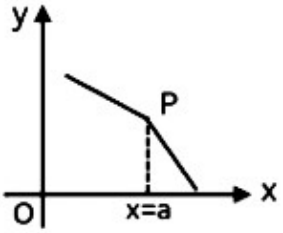
Note :

It should be noted that we can talk of monotonicity of f(x) at x = a only if x = a lies in the domain of f(x), without any consideration of continuity or differentiability of f(x) at x = a.



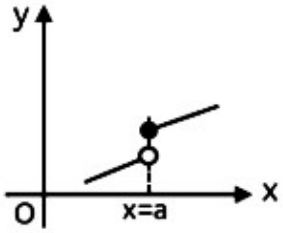
Increasing at x=a

(a)



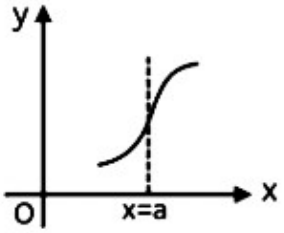
decreasing at x=a

(b)



Increasing at x=a

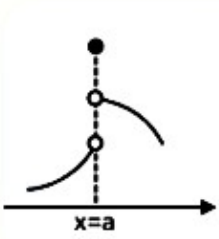
(c)



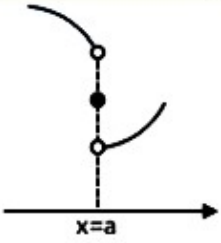
Increasing at x=a

(d)

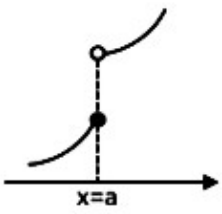
Ex. **For each of the following graph, comment whether f(x) is increasing or decreasing or neither increasing nor decreasing at x = a.**



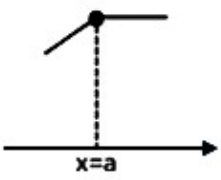
(A)



(B)



(C)



(D)

Sol. **(A)** Neither monotonically increasing nor decreasing as $f(a-h) < f(a)$ and $f(a+h) < f(a)$

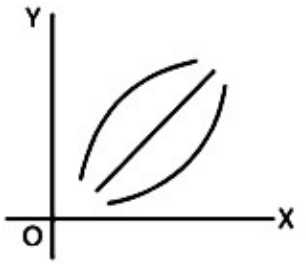
(B) Monotonically decreasing as $f(a-h) > f(a) > f(a+h)$

(C) Monotonically increasing as $f(a-h) < f(a) < f(a+h)$

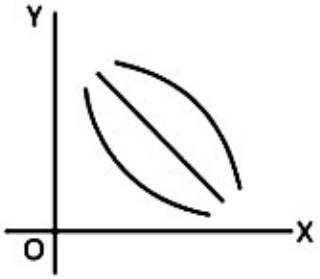
(D) Neither monotonically increasing nor decreasing as $f(a-h) < f(a)$ but $f(a+h) = f(a)$

MONOTONICITY IN AN INTERVAL :

(a) For an increasing function in some interval, if $\Delta x > 0 \Leftrightarrow \Delta y > 0$ or $\Delta x < 0 \Leftrightarrow \Delta y < 0$ then f is said to be monotonic (strictly) increasing in that interval. In other words if Δy and Δx have the same sign i.e. $\frac{dy}{dx} > 0$ for increasing function. Hence if $\frac{dy}{dx} > 0$ in some (interval) then y is said to be increasing function in that interval and conversely if f(x) is increasing in some interval then $\frac{dy}{dx} > 0$ in that Interval.



(b) Similarly, if $\frac{dy}{dx} < 0$ in some interval then y is decreasing in that interval and vice-versa.

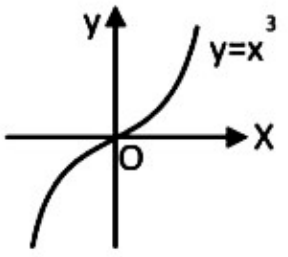


Note :

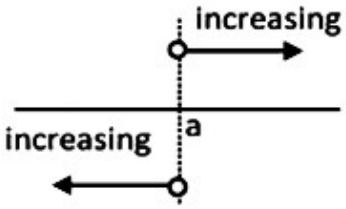
Hence to find the intervals of monotonicity for a function $y = f(x)$ one has to find $\frac{dy}{dx}$ and solve the inequality, $\frac{dy}{dx} > 0$ or $\frac{dy}{dx} < 0$ The solution of this inequality gives the interval of monotonicity.

It should however be noted that

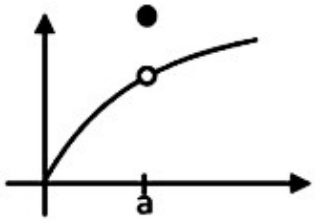
(a) $\frac{dy}{dx}$ at some point may be equal to zero but f(x) may still be increasing at x = a. Consider $f(x) = x^3$ which is increasing at x = 0 although



$f'(x) = 0$. This is because $f(0 + h) > f(0)$ and $f(0 - h) < f(0)$. At all such points where $\frac{dy}{dx} = 0$ but y is still increasing or decreasing are known as point of inflection, which indicate the change of concavity of the curve.

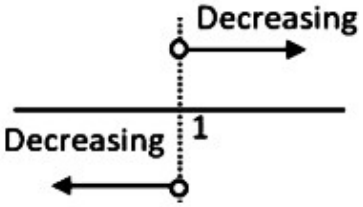


(b) If f is increasing for $x > a$ and f is also increasing for $x < a$ then f is also increasing at $x = a$ provided $f(x)$ is continuous at $x = a$.



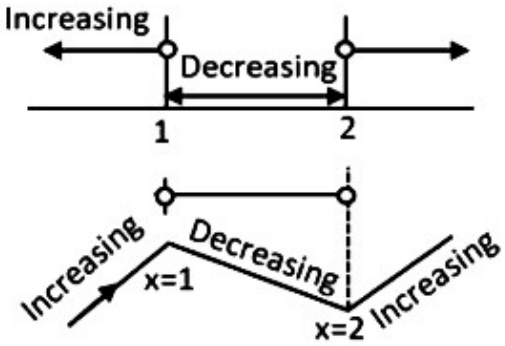
(c) If $f(x)$ is discontinuous at $x = a$ then one can draw the graph as shown $x = a$ is the point of maxima

(d) Similarly if f is decreasing for $x > a$ and f is also decreasing for $x < a$ then f is also decreasing at $x = a$ provided $f(x)$ is continuous at $x = a$.



(e) However if $f(a)$ is not defined then monotonicity will not be indicated at $x = a$

e.g. $f(x) = \frac{1}{x-1}$ is decreasing for $x \in (-\infty, 1) \cup (1, \infty)$. However if f is increasing and decreasing as shown then at $x = 1$ and $x = 2$, f will have extremum values, being maximum at $x = 1$ and minimum at $x = 2$.



Increasing and decreasing functions :

(i) A function $f(x)$ is said to be monotonically increasing for all such interval (a, b) where $f'(x) \geq 0$ and equality may hold only for discrete values of x, i.e., $f'(x)$ does not identically become zero for $x \in (a, b)$ or any sub-interval.

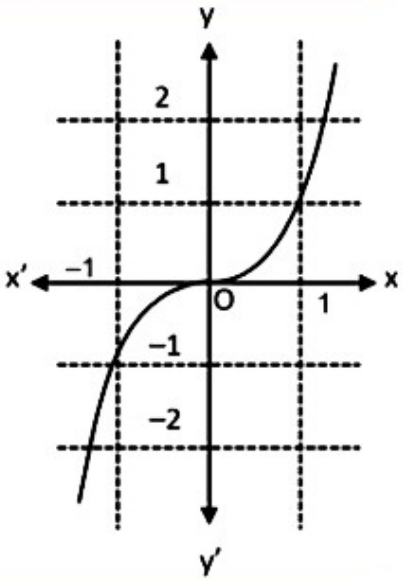
(ii) $f(x)$ is said to be monotonically decreasing for all such interval (a, b) where $f'(x) \leq 0$ and equality may hold only for discrete values of x.

Note :

By discrete points, we mean those points where f'(x) = 0 does not form any interval.

Ex. **Prove that $f(x) = x^3$ is an increasing function.**

Sol. Clearly $f'(x) = 3x^2 \geq 0$ in $(-\infty, \infty)$ and equality holds only at $x = 0$ and not in any interval, therefore $f(x) = x^3$ is an increasing function in $(-\infty, \infty)$.



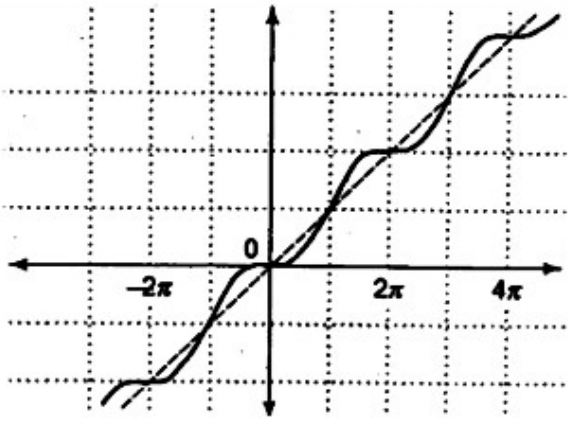
Ex. **Prove that $f(x)=x - \sin x$ is an increasing function.**

Sol. $f(x) = x - \sin x$

$\Rightarrow \quad f'(x) = 1 - \cos x$

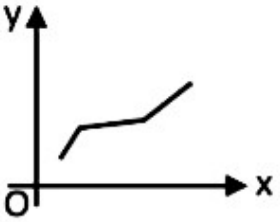
Now, $f'(x) > 0$ everywhere except at $x = 0, \pm 2\pi, \pm 4\pi$ etc. but all these points are discrete and do not form an interval hence we can conclude that $f(x)$ is monotonically increasing for $x \in R$

In fact, we can also see it graphically.



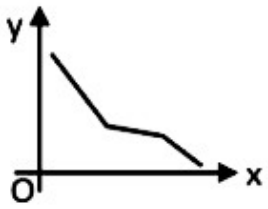
Non-decreasing function :

$f(x)$ is said to be non-decreasing in domain if for every $x_1, x_2 \in D_1$, $x_1 > x_2 \Rightarrow f(x_1) \geq f(x_2)$. It means that the value of $f(x)$ would never decrease with an increase in the value of x (Figure).



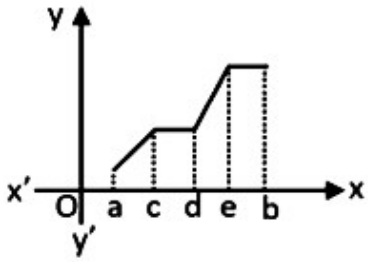
Non-Increasing function :

f(x) is said to be non-increasing in domain if for every $x_1, x_2 \in D_f$, $x_1 > x_2$ $f(x_1) \le f(x_2)$. It means that the value of f(x) would never increase with an increase in the value of x (Figure).



Let us consider another function whose graph is shown for $x \in (a,b)$

Here also $f'(x) \ge 0$ for all $x \in (a, b)$ but note that in this case equality of $f'(x) = 0$ holds for all $x \in (c, d)$ and (e, b). Here $f'(x)$ becomes identically zero and hence the given function cannot be assumed to be monotonically increasing for $x \in (a, b)$.



Note :

(i) If a function is monotonic at x = a it can not have extremum point at x = a and vice versa i.e. a point on the curve can not simultaneously be an extremum as well as monotonic point.

(ii) If f is increasing then nothing definite can be said about the function f'(x) w.r.t. its increasing or decreasing behaviour.

Ex. **Find intervals of monotonicity of $f(x) = \frac{x}{\ln x}$**

Sol. $f'(x) = \frac{\ln x - 1}{(\ln x)^2}$

sign of f'(x) $\overset{\text{- - - - - - - - - - + + + + +}}{\underset{0 \quad 1 \quad e}{\longrightarrow}}$

strictly increasing in (e, ∞) and strictly decreasing in $(0, 1) \cup (1, e)$

Ex. **If the function $f(x) = (a + 2)x^3 - 3ax^2 + 9ax - 1$ is always decreasing $\forall x \in R$, find 'a'**

Sol. $f'(x) = 3(a + 2)x^2 - 6ax + 9a \le 0 \ \forall \ x \in R$

$\Rightarrow 3(a + 2) < 0 \quad \& \quad 36a^2 - 4 \cdot 3(a + 2) \cdot 9a \le 0$

$\Rightarrow a < -2 \quad \& \quad a^2 - 3a(a + 2) \le 0$

$\Rightarrow a <- 2 \ ...(1) \quad \& \quad a^2 + 3a \ge 0$

$\Rightarrow a \in (-\infty, -3] \cup [0, \infty)$ ...(2)

from (1) and (2)

$a \leq -3$ **Ans.**

Ex. **Find intervals of monotonicity of following functions :**

(a) $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 7$ **(b)** $f(x) = \dfrac{2x}{1+x^2}$

(c) $f(x) = \ln(x^2 - 2x)$ **(d)** $f(x) = \dfrac{|x-1|}{x^2}$

Sol.

(a) We have $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 7$, $x \in R$

and $f'(x) = 4x^3 - 24x^2 + 44x - 24$

$= 4(x-1)(x-2)(x-3)$

From the sign scheme for f' (x), we can see that f(x)

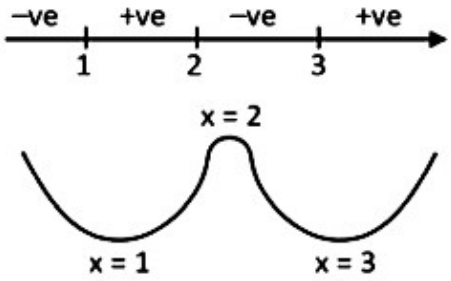
strictly decreases in $(-\infty, 1)$

strictly increases in $(1, 2)$

strictly decreases in $(2, 3)$

strictly increases in $(3, \infty)$.

The shape of the curve is drawn alongside.



(b) We have $f(x) = \dfrac{2x}{1+x^2}, x \in R$

$$f'(x) = \frac{(1+x^2)2 - 2x(2x)}{(1+x^2)^2} = \frac{-2(x^2-1)}{(1+x^2)^2} = \frac{-2(x+1)(x-1)}{(1+x^2)^2}$$

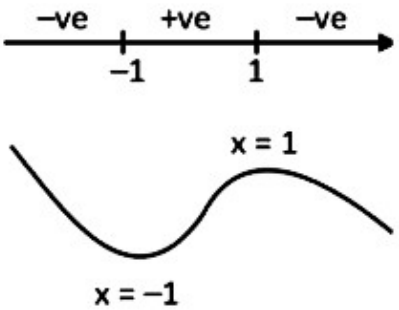
From the sign scheme for f' (x), we can see that f(x)

strictly decreases in $(-\infty, -1)$

strictly increases in $(-1, 1)$

strictly decreases in $(1, \infty)$

The shape of the curve is shown alongside.



(c) We have $f(x) = \ln(x^2 - 2x)$, $x \in (-\infty, 0) \cup (2, \infty)$

and $f'(x) = \dfrac{2x-2}{x^2-2x} = \dfrac{2(x-1)}{x(x-2)}$

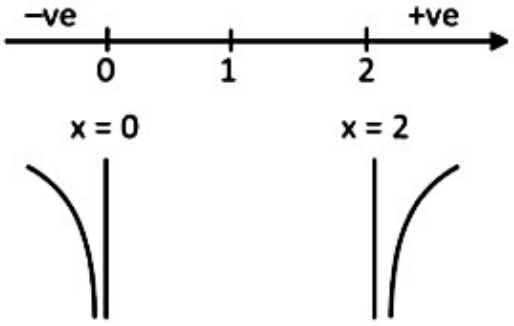
From the sign scheme for f' (x), we can see that f(x)

strictly decreases in $(-\infty, 0)$

strictly increases in $(2, \infty)$.

Also, we can see that $f(0^-) = -\infty$

and $f(2^+) = -\infty$.



(d) we have $f(x) = -\frac{(x-1)}{x^2}, x < 1$ and $f(x) = \frac{x-1}{x^2}, x \geq 1$

And $f'(x) = \frac{-2}{x^3} + \frac{1}{x^2} = \frac{x-2}{x^3}, x < 1$ and $f'(x) = \frac{2-x}{x^3}, x > 1$

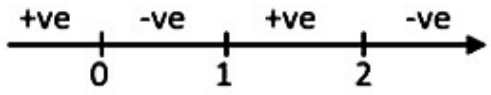
Now, from the sign scheme for f'(x), we have

$\Rightarrow$ f(x) strictly increases in $(-\infty, 0)$

strictly decreases in (0, 1)

strictly increases in (1, 2)

strictly decreases in $(2, \infty)$.



Ex. **If $\varphi(x) = f(x) + f(1-x)$ and $f''(x) < 0$ in $(-1, 1)$, then show that $\varphi(x)$ strictly increases in $\left(0, \frac{1}{2}\right)$.**

Sol. We have $\varphi(x) = f(x) + f(1-x)$ and $\varphi'(x) = f'(x) - f'(1-x)$

Which vanishes at points given by $x = 1 - x$ i.e. $x = \frac{1}{2}$

$f''(x) < 0$ so $f'(x)$ is decreasing for $x \in \left(0, \frac{1}{2}\right)$ i.e. $1 - x > x \Rightarrow f'(1-x) < f'(x)$

hence $\varphi'(x) > 0 \ \forall \ x \in \left(0, \frac{1}{2}\right)$

Hence, $\varphi(x)$ strictly increases in $\left(0, \frac{1}{2}\right)$.

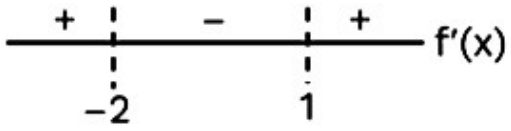
Q1 **Find the interval in which $f(x) = 2x^3 + 3x^2 - 12x - 1$ is increasing.**

Sol. $f'(x) = 6x^2 + 6x - 12$

$= 6(x-1)(x+2)$

$\Rightarrow f'(x) \geq 0 \ \forall x \in (-\infty, -2] \cup [1, \infty)$

$\Rightarrow f(x)$ is increasing in $(-\infty, -2] \cup [1, \infty)$

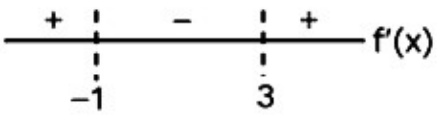


Q2 **Find the interval in which $f(x) = x^3 - 3x^2 - 9x + 20$ is strictly increasing or strictly decreasing.**

Sol. $f'(x) = 3x^2 - 6x - 9$

$= 3(x^2 - 2x - 3)$

$= 3(x+1)(x-3)$



$\Rightarrow f'(x) > 0 \forall x \in (-\infty, -1) \cup (3, \infty)$

$\Rightarrow f'(x) < 0 \forall x \in (-1, 3)$.

$\Rightarrow f(x)$ is strictly increasing in $(-\infty, -1) \cup (3, \infty)$

$\Rightarrow f(x)$ is strictly decreasing in $(-1, 3)$

Q3 **Prove that the following functions are increasing for the given intervals :**

(a) $f(x) = e^x + \sin x, x \in R^+$

(b) $f(x) = \sin x + \tan x - 2x, x \in \left(0, \frac{\pi}{2}\right)$

(c) $f(x) = \sec x - \operatorname{cosec} x, x \in \left(0, \frac{\pi}{2}\right)$

Sol. **(a)** $f'(x) = \underbrace{e^x}_{\geq 1} + \underbrace{\cos x}_{> -1} \quad \forall x \in R^+$

$\Rightarrow f'(x) > 0 \forall x \in R^+$

$\Rightarrow f(x)$ is increasing

(b) $f'(x) = \cos x + \sec^2 x - 2$

$\Rightarrow f'(x) > \cos^2 x + \sec^2 x - 2 \qquad \left(\because \cos x > \cos^2 x \ \forall \ x \in \left(0, \frac{\pi}{2}\right)\right)$

$\Rightarrow f'(x) > 0 \qquad \left(\because \cos^2 x + \sec^2 x > 2\right)$

$\Rightarrow f(x)$ is increasing for $x \in \left(0, \frac{\pi}{2}\right)$.

(c) $f'(x) = \sec x \tan x + \operatorname{cosec} x \cot x > 0 \forall \ x \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow f(x)$ is increasing in $\left(0, \frac{\pi}{2}\right)$ **Hence Proved.**

Q4 **If $f(x) = xe^{x(1-x)}$, then f(x) is**

(A) increasing on $\left(\frac{-1}{2}, 1\right)$ **(B) decreasing on** $\left[\frac{-1}{2}, 1\right]$

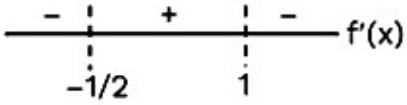
(C) increasing on R **(D) decreasing on R**

Sol. **(A)**

$$f'(x) = e^{x(1-x)} + x \cdot e^{x(1-x)}(1-2x)$$

$$= e^{x(1-x)}(1+x-2x^2)$$

$$= -e^{x(1-x)}(x-1)(2x+1)$$



$\Rightarrow f'(x) > 0 \; \forall \; x \in \left(\frac{-1}{2}, 1\right)$

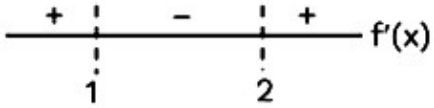
$\Rightarrow f(x)$ is increasing on $\left(\frac{-1}{2}, 1\right)$.

Q5 **Let $f(x) = \int e^x (x-1)(x-2)dx$, then f(x) decreases in interval.**

(A) $(-\infty, 2)$ **(B) (–2,–1)** **(C) (1, 2)** **(D)** $(2, \infty)$

Sol. **(C)**

$$f'(x) = e^x (x-1)(x-2)$$



$\Rightarrow f'(x) < 0$ in (1, 2)

$\Rightarrow f(x)$ decreases in (1, 2)

Q6 **Separate the intervals of monotonicity of following functions :**

(A) $f(x) = 3x^4 - 8x^3 - 6x^2 + 24x + 7$

(B) $f(x) = -\sin^3 x + 3\sin^2 x + 5, x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

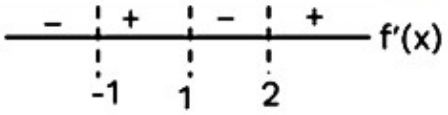
(C) $f(x) = (2^x - 1)(2^x - 2)^2$

(D) $f(x) = \cos\left(\frac{\pi}{x}\right)$

Sol. **(A)** $f'(x) = 12x^3 - 24x^2 - 12x + 24$

$$= 12(x^3 - 2x^2 - x + 2)$$

$$= 12(x-1)(x+1)(x-2)$$



$\Rightarrow f(x)$ is increasing in $[-1, 1] \cup [2, \infty)$ and decreasing in $(-\infty, -1) \cup (1, 2)$.

(B) $f'(x) = -3\sin^2 x\cos x + 6\sin x\cos x$

$= -3\sin x\cos x(\sin x - 2)$

$= 3\sin x\cos x(2 - \sin x)$

$2 - \sin x > 0 \ \forall \ x \in R$

$\cos x \geq 0 \forall x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

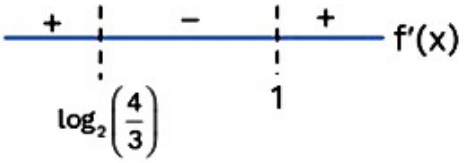
$\sin x \geq 0 \forall x \in \left[0, \frac{\pi}{2}\right]$

$\Rightarrow f'(x) \geq 0 \forall x \in \left[0, \frac{\pi}{2}\right]$ and $f'(x) \leq 0 \ \forall x \in \left[-\frac{\pi}{2}, 0\right]$

$\Rightarrow f(x)$ is increasing in $\left[0, \frac{\pi}{2}\right]$ and decreasing in $\left[-\frac{\pi}{2}, 0\right]$

(C) $f'(x) = 2^x \ell n(2)(2^x - 2)^2 + (2^x - 1)2(2^x - 2)2^x \ell n(2)$

$= 2^x \ell n(2)(2^x - 2)(3 \cdot 2^x - 4)$



$\Rightarrow f'(x) = 0$ at $x = 1$ and $\log_2\left(\frac{4}{3}\right)$

$\Rightarrow f(x)$ is increasing in $\left(-\infty, \log_2\left(\frac{4}{3}\right)\right) \cup (1, \infty)$ and f(x) is decreasing in $\left(\log_2\left(\frac{4}{3}\right), 1\right)$.

(D) $f'(x) = -\sin\left(\frac{\pi}{x}\right)\left(\frac{-\pi}{x^2}\right)$

$= \left(\frac{\pi}{x^2}\right)\sin\left(\frac{\pi}{x}\right)$

$\Rightarrow f'(x) > 0$ when $\frac{\pi}{x} \in (2n\pi, (2n+1)\pi), n \in I$

$f'(x) > 0$ when $x \in \left(\frac{1}{2n+1}, \frac{1}{2n}\right), n \in I$

$\Rightarrow f(x)$ increases for $x \in \left(\frac{1}{2n+1}, \frac{1}{2n}\right), n \in I$ and it decreases else everywhere.

Q7 **Find the complete set of values of λ, for which the function**

$$f(x)=\begin{cases} x+1 & ,x<1 \\ \lambda & ,x=1 \\ x^2-x+3, & x>1 \end{cases}$$ **is increasing at x = 1.**

Sol. For f(x) to be increasing at x = 1,

$$f(1-h)\le f(1)\le f(1+h), h\to 0^+$$

$$\Rightarrow 2-h\le \lambda \le h+h^2+3, h\to 0^+$$

$$\Rightarrow 2\le \lambda \le 3$$

Q8 **If $f:[0,\infty)\to R$ is the function defined by $f(x)=\frac{e^{x^2}-e^{-x^2}}{e^{x^2}+e^{-x^2}}$, then check whether f(x) is injective or not.**

Sol.

$$f'(x)=\frac{\left(e^{x^2}+e^{-x^2}\right)\left(e^{x^2}(2x)+e^{-x^2}(2x)\right)-\left(e^{x^2}-e^{-x^2}\right)2x\left(e^{x^2}-e^{-x^2}\right)}{\left(e^{x^2}+e^{-x^2}\right)^2}$$

$$=\frac{2x\left[\left(e^{x^2}+e^{-x^2}\right)^2-\left(e^{x^2}-e^{-x^2}\right)^2\right]}{\left(e^{x^2}+e^{-x^2}\right)^2}$$

$$=\frac{8x}{\left(e^{x^2}+e^{-x^2}\right)^2}\ge 0\ \forall\ x\in[0,\infty)$$

$\therefore f(x)$ is increasing in $[0,\infty)$

$\Rightarrow f(x)$ is injective.

Q9 **Let f(x) and g(x) be two continuous functions defined from R→R, such that $f(x_1)>f(x_2)$ and $g(x_1)<g(x_2)\ \forall\ x_1>x_2$. Find the solution set of**

$$f\left(g\left(\alpha^2-2\alpha\right)\right)>f\left(g(3\alpha-4)\right)$$

Sol. $f(x_1)>f(x_2)\forall x_1>x_2 \Rightarrow f$ is increasing function.

$g(x_1)<g(x_2)\forall x_1>x_2 \Rightarrow g$ is a decreasing function.

$$\therefore f\left(g\left(\alpha^2-2\alpha\right)\right)>f\left(g(3\alpha-4)\right)$$

$\Rightarrow g(\alpha^2 - 2\alpha) > g(3\alpha - 4)$ ($\because$ f is increasing)

$\Rightarrow \alpha^2 - 2\alpha < 3\alpha - 4$ ($\because$ g is decreasing)

$\Rightarrow \alpha^2 - 5\alpha + 4 < 0$

$\Rightarrow \alpha \in (1,4)$.

Q10 **Find the number of solution of the equation** $3\tan x + x^3 = 2$ **in** $\left(0, \frac{\pi}{4}\right)$

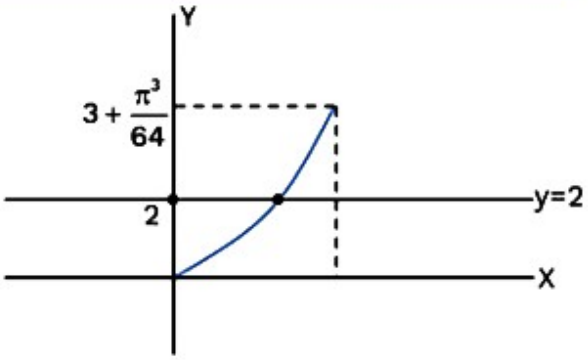
Sol. Let $f(x) = 3\tan x + x^3$

$\Rightarrow f'(x) = 3\sec^2 x + 3x^2 > 0$ in $\left(0, \frac{\pi}{4}\right)$

$\Rightarrow$ So f(x) is increasing and one–one in $\left(0, \frac{\pi}{4}\right)$

$f(0) = 0$ and $f\left(\frac{\pi}{4}\right) = 3 + \frac{\pi^3}{64} > 2$.

From graph, it is clear that there is only 1 solution.



GREATEST AND LEAST VALUE OF A FUNCTION

Case-I:

If a continuous function y = f(x) is strictly increasing in the closed interval [a, b] then f(a) is the least value. (figure-1) and f(b) is greatest value

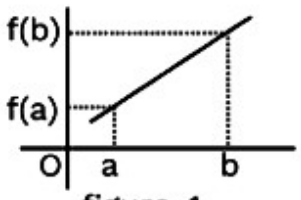


figure-1

Case-II:

If a continuous function f(x) is strictly decreasing in [a, b] then f(b) is the least and f(a) is the greatest value of f(x) in [a, b]. (figure-2)

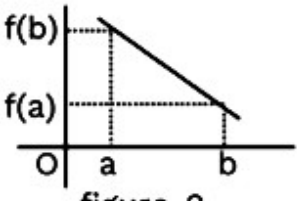


figure-2

Case-III:

However if f(x) is non monotonic in [a, b] and is continuous then the greatest and least value of f(x) in [a, b] are those where f'(x) =0 or f'(x) does not exist or at the extreme values. (figure-3)

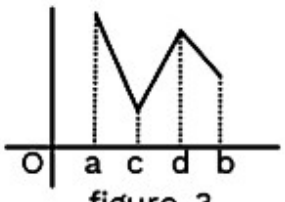


figure-3

Ex. **Find least and greatest value of $f(x) = e^{x^2-4x+3}$ in $[-5, 5]$**

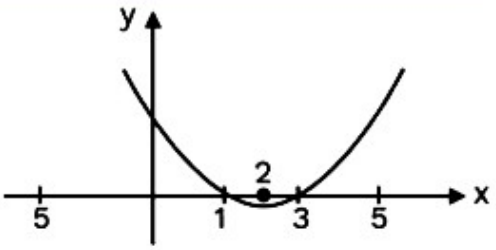
Sol. $f(x) = e^{x^2-4x+3}$

for f(x) max → $x^2 - 4x + 3$ be maximum in $[-5, 5]$

$x^2 - 4x + 3$ will be maximum at $x = -5$ in the given interval.

i.e., $25 + 20 + 3 = 48$

$\therefore$ Max $f(x) = e^{48}$ at $x = -5$



$x^2 - 4x + 3$ will be minimum at $x = 2$ i.e., $4 - 8 + 3 = -1$

$\therefore$ Min $f(x) = e^{-1}$ at $x = 2$.

Ex. **Find the image of interval $[-1, 3]$ under the mapping specified by the function $f(x) = 4x^3 - 12x$.**

Sol. $f'(x) = 12x^2 - 12 = 12(x^2 - 1)$

$f'(x) = 0$ at $x = \pm 1$, $f(-1) = 8$, $f(1) = -8$

$f(3) = 72 \Rightarrow$ greatest value is 72 and least value is -8

Thus the image of interval $[-1,3]$ will be $[-8,72]$

Ex. **Find the range of the following function $f(x) = \sqrt{x-3} + 2\sqrt{5-x}$**

Sol. We have $f(x) = \sqrt{x-3} + 2\sqrt{5-x}$

Whose domain is $x \in [3, 5]$ and its derivative is

$$f'(x) = \frac{1}{2\sqrt{x-3}} - \frac{1}{\sqrt{5-x}} = \frac{\sqrt{5-x} - 2\sqrt{x-3}}{2\sqrt{x-3}\sqrt{5-x}}$$

Now, solving

$$\sqrt{5-x} > 2\sqrt{x-3} \quad \text{i.e.} \quad 5 - x > 4(x-3) \text{ given } x < \frac{17}{5}$$

Hence, we have

$$f'(x) > 0 \forall x \in \left(3, \frac{17}{5}\right) \text{ and } f'(x) < 0 \forall x \in \left(\frac{17}{5}, 5\right)$$

$\Rightarrow$ f(x) strictly increases in $\left(3, \frac{17}{5}\right)$ and strictly decreases in $\left(\frac{17}{5}, 5\right)$.

Now, we have

$$f(3) = 2\sqrt{2}, f(5) = \sqrt{2} \text{ and } f\left(\frac{17}{5}\right) = \sqrt{\frac{17}{5} - 3} + 2\sqrt{5 - \frac{17}{5}} = \sqrt{10}.$$

Hence, the range is $y \in [\sqrt{2}, \sqrt{10}]$

Ex. **Find the range of the following function** $f(x)=\dfrac{x^4-x^2-2x+8}{x^4-x^2-2x+4}$

Sol. We have

$$f(x)=\frac{x^4-x^2-2x+8}{x^4-x^2-2x+4}=1+\frac{4}{x^4-x^2-2x+4}=1+\frac{4}{\left(x^2-1\right)^2+(x-1)^2+2}$$

Let $g(x)=(x^2-1)^2+(x-1)^2+2$, whose least value = 2
and greatest value = ∞

Thus, we have for f(x) greatest value = $1+\dfrac{4}{2}=3$ and least value = $1+\dfrac{4}{\infty}=1$.

Also, f(x) is continuous and defined on R. Hence, the range of f(x), is $y \in (1, 3]$.

Q1 **Find range of following functions :**

(a) $f(x)=x^2e^{-x}$

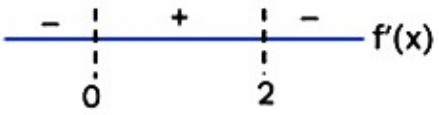
(b) $f(x)=x+\ell n(1-4x)$

(c) $f(x)=\dfrac{x}{\ell n(x)}$

(d) $f(x)=2x^2-\ell n|x|$

Sol. **(a)** $f'(x)=2xe^{-x}-x^2e^{-x}$

$=-e^{-x}x(x-2)$



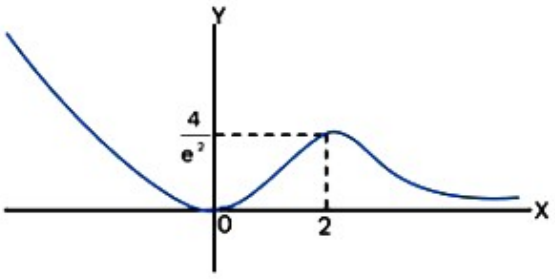
$\Rightarrow f(x)$ is decreasing in $(-\infty,0)\cup(2,\infty)$ and f(x) is increasing in (0, 2).

$f(-\infty)=+\infty$; $f(0)=0$;

$f(2)=\dfrac{4}{e^2}$; $f(\infty)\approx 0^+$

Graph :

$\Rightarrow$ From graph, it is clear that range of f(x) is $[0,\infty)$

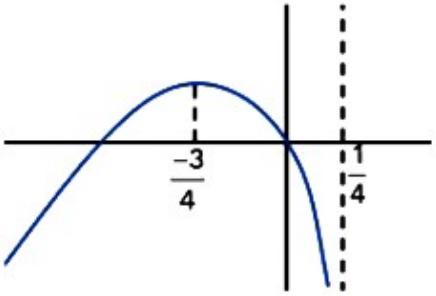


(b) $1-4x>0 \Rightarrow$ Domain : $\left(-\infty,\dfrac{1}{4}\right)$

$f'(x)=1-\dfrac{4}{1-4x}=\dfrac{-(4x+3)}{1-4x}$

$\Rightarrow f'(x)<0\ \forall\ x\in\left(-\dfrac{3}{4},\dfrac{1}{4}\right)$

and $f'(x)>0\ \forall\ x\in\left(-\infty,\dfrac{-3}{4}\right)$.



$\Rightarrow f(x)$ attains maxima at $x = \frac{-3}{4}$

$\lim\limits_{x\to-\infty} f(x) = -\infty$

$f\left(\frac{-3}{4}\right) = \frac{-3}{4} + \ln 4 > 0$

$\lim\limits_{x\to\left(\frac{1}{4}\right)^-} f(x) = \frac{1}{4} - \infty = -\infty$

$\Rightarrow$ Range of f(x) is $\left(-\infty, \ell n(4) - \frac{3}{4}\right]$.

(c) Domain: $x > 0, x \neq 1$

$f'(x) = \frac{\ell nx - 1}{(\ell nx)^2}$

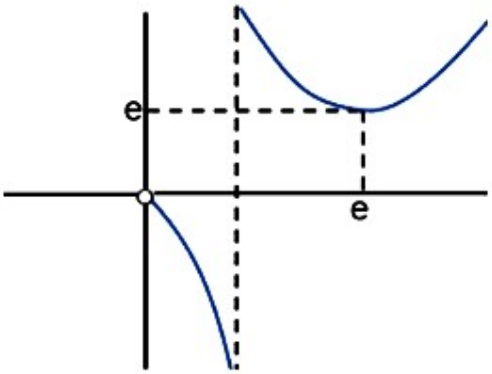
$\Rightarrow f'(x) > 0 \ \forall x > e$ & $f'(x) < 0 \ \forall x \in (0,1) \cup (1,e)$

$\Rightarrow f(x)$ is decreasing in $(0,1) \cup (1,e)$ and increasing in (e,∞).

$\lim\limits_{x\to0^+} f(x) = 0^-$; $\lim\limits_{x\to1^-} f(x) = -\infty$

$\lim\limits_{x\to1^+} f(x) = +\infty$; $f(e) = e$; $f(\infty) = \infty$

$\Rightarrow$ Range of f(x) is $(-\infty, 0) \cup [e, \infty)$



(d) Domain : $x \in R - \{0\}$

f(x) is also an even function.

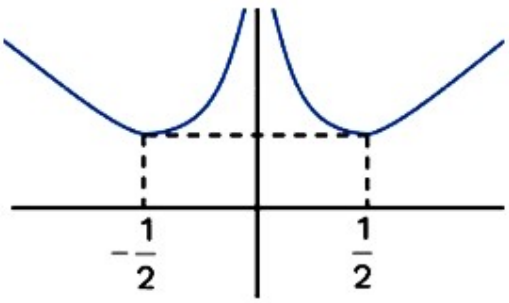
For $x > 0$, $f'(x) = 4x - \frac{1}{x} = \frac{(2x-1)(2x+1)}{x}$

$\Rightarrow f'(x) > 0 \ \forall x > \frac{1}{2}$ & $f'(x) < 0 \ \forall x \in \left(0, \frac{1}{2}\right)$

$\Rightarrow f(x)$ is decreasing in $\left(0, \frac{1}{2}\right)$ and increasing in $\left(\frac{1}{2}, \infty\right)$.

$f(0^+) = +\infty$; $f\left(\frac{1}{2}\right) = \frac{1}{2} + \ell n(2)$; $f(\infty) = \infty$

$\Rightarrow$ Range of f(x) is $\left[\frac{1}{2} + \ell n2, \infty\right)$



Q2 **$f(x) = ax - \sin x$. Find 'a' if the function has no critical point.**

Sol. $f'(x) = a - \cos x \neq 0$

$\Rightarrow a \neq \cos x \quad \forall\ x \in R$

$\Rightarrow a \in (-\infty, -1) \cup (1, \infty)$

Q3 **If the function $f(x) = \sin x - a\sin 2x - \frac{1}{3}\sin 3x + 2ax$ is monotonic increasing $\forall x \in R$, then find 'a'.**

Sol. $f'(x) = \cos x - 2a\cos 2x - \cos 3x + 2a \geq 0$

$\Rightarrow 2\sin x \sin 2x + 2a\left(2\sin^2 x\right) \geq 0$

$\Rightarrow 4\sin^2 x \cos x + 4a\sin^2 x \geq 0$

$\Rightarrow 4\sin^2 x(\cos x + a) \geq 0$

$\Rightarrow \cos x + a \geq 0$

$\Rightarrow a \geq -\cos x \quad \forall x \in R$

$\Rightarrow a \geq 1$

Q4 **Prove that $f(x) = \int_{x^2}^{x^3} \frac{dt}{\ell nt}, (x > 0)$ is always an increasing function of x. (Note that f(0) and f(1) is not defined.)**

Sol. $f'(x) = \frac{1}{\ell n(x^3)}(3x^2) - \frac{1}{\ell n(x^2)}(2x)$

$= \frac{x^2 - x}{\ell nx} = \frac{x(x-1)}{\ell nx}$

$\Rightarrow f'(x) > 0 \quad \forall\ x > 0, x \neq 1.$

$\therefore$ f(x) is an increasing function of x. **Hence Proved.**

Q5 **Find all possible 'b' for which $f(x) = \sin 2x - 8(b+2)\cos x - (4b^2 + 16b + 6)x$ is monotonic decreasing $\forall x \in R$ and has no critical point.**

Sol. $f'(x) \neq 0$ and $f'(x) \leq 0$

$\Rightarrow f'(x) < 0$

$$\Rightarrow f'(x) = 2\cos 2x + 8(b+2)\sin x - \left(4b^2 + 16b + 6\right) < 0$$

$$\Rightarrow 2\left(1 - 2\sin^2 x\right) + 8(b+2)\sin x - \left(4b^2 + 16b + 6\right) < 0$$

$$\Rightarrow 4\left(\sin^2 x - 2(b+2)\sin x + b^2 + 4b + 1\right) > 0$$

$$\Rightarrow \left(\sin x - (b+2)\right)^2 > 3$$

$$\Rightarrow \sin x - b - 2 < -\sqrt{3} \text{ or } \sin x - b - 2 > \sqrt{3}$$

$$\Rightarrow b > \sin x + \sqrt{3} - 2 \text{ or } b < \sin x - \sqrt{3} - 2$$

$$\Rightarrow b > 1 + \sqrt{3} - 2 \text{ or } b < -1 - \sqrt{3} - 2$$

$$\Rightarrow b \in \left(-\infty, -3 - \sqrt{3}\right) \cup \left(\sqrt{3} - 1, \infty\right)$$

Q6 $\mathbf{y = \int_{\frac{5\pi}{4}}^{x} (3\sin t + 4\cos t)\,dt}$ **in** $\mathbf{\left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]}$**. Find the minimum and maximum value of y.**

Sol. $\frac{dy}{dx} = 3\sin x + 4\cos x < 0$ in $\left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$

$\Rightarrow$ y is decreasing in this interval.

So $y_{min} = f\left(\frac{4\pi}{3}\right)$

$$= \int_{\frac{5\pi}{4}}^{\frac{4\pi}{3}} (3\sin t + 4\cos t)\,dt$$

$$= -3\cos t + 4\sin t \Big]_{\frac{5\pi}{4}}^{\frac{4\pi}{3}}$$

$$= \left(\frac{3}{2} - 2\sqrt{3}\right) - \left(\frac{3}{\sqrt{2}} - 2\sqrt{2}\right)$$

$$\Rightarrow y_{min} = \frac{3}{2} + \frac{1}{\sqrt{2}} - 2\sqrt{3}$$

$$y_{max} = f\left(\frac{5\pi}{4}\right) = 0$$

Q7 **Use the function $f(x)=x^{\frac{1}{x}}\,(x>0)$ to ascertain whether π^e or e^π is greater.**

Sol. $f(x)=x^{\frac{1}{x}}=e^{\frac{\ell n(x)}{x}}$

$\Rightarrow f'(x)=e^{\left(\frac{\ell nx}{x}\right)}\left(\frac{1-\ell nx}{x^2}\right)$

$\Rightarrow f'(x)>0 \Rightarrow 1-\ell n(x)>0$

$\Rightarrow x\in(0,e)$

$\Rightarrow f(x)<0 \forall x\in(e,\infty)$

So f(x) increases in (0, e) and then decreases.

Since $\pi>e, \pi^{\frac{1}{\pi}}<e^{\frac{1}{e}}$

$\Rightarrow\left(\pi^{\frac{1}{\pi}}\right)^{\pi e}<\left(e^{\frac{1}{e}}\right)^{\pi e}$

$\Rightarrow \pi^e<e^\pi$

Q8 **If $f(x)=|px-q|+r|x|, x\in R, p,q,r>0$, assume its minimum value only at one point, then find the relation between p, q, r.**

Sol. $$f(x)=\begin{cases} px-q+rx; & x\geq\frac{q}{p} \\ q-px+rx; & 0\leq x<\frac{q}{p} \\ q-px-rx; & x<0 \end{cases}$$

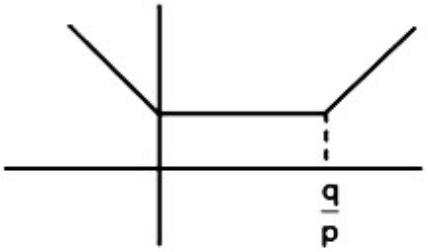
$$\Rightarrow f'(x)=\begin{cases} p+r; & x\geq\frac{q}{p} \\ r-p; & 0\leq x<\frac{q}{p} \\ -p-r; & x<0 \end{cases}$$

$\Rightarrow f'(x)>0 \quad \forall x\geq\frac{q}{p}$

$\Rightarrow f'(x) < 0 \quad \forall x < 0$

In $0 \le x \le \frac{q}{p}$,

Case-I : $r = p \Rightarrow f'(x) = 0$



Graph will have minimum $\forall x \in \left(0, \frac{q}{p}\right)$

So this case is rejected

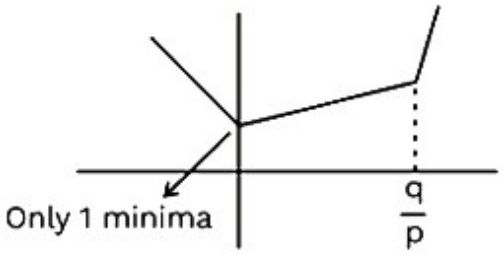
Graph :

$\Rightarrow r \neq p$

Case-II : $r > p$

$\Rightarrow f'(x) > 0$

Graph:

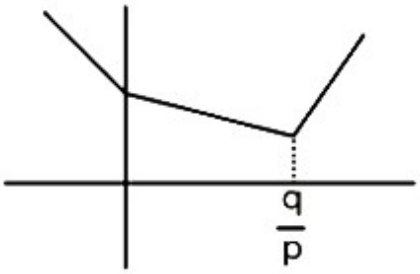


$\Rightarrow r > p$ satisfies the condition

Case-III : $r < p$

$\Rightarrow f'(x) < 0$

Graph :



$\Rightarrow$ Again, f(x) has minima at only $\frac{q}{p}$.

So, $r < p$ satisfies the condition.

$\Rightarrow$ Overall, for f(x) to have only one minima, necessary condition is $p \neq r$.

Q9 **Find the minimum value of the function, $f(x) = 8^x + 8^{-x} - 4\left(4^x + 4^{-x}\right), \forall x \in R$.**

Sol. Let $2^x + 2^{-x} = t$

$\Rightarrow t^2 = 4^x + 4^{-x} + 2$

$\Rightarrow t^3 = 8^x + 8^{-x} + 3\left(2^x + 2^{-x}\right)$

Now, Since $2^x + 2^{-x} \in [2, \infty) \Rightarrow t \in [2, \infty)$

$\Rightarrow f(x) = 8^x + 8^{-x} - 4\left(4^x + 4^{-x}\right)$

$= \left(t^3 - 3t\right) - 4\left(t^2 - 2\right), t \in [2, \infty)$

Differentiating, we get $3t - 8t - 3$

$= (t - 3)(3t + 1) > 0$ when $t > 3$

$\therefore g(t) = t^3 - 4t^2 - 3t + 8$ decreases in [2, 3] and then increases.

$\Rightarrow g(t)$ has minima at $t = 3$.

$\Rightarrow$ minima value $= g(3) = 27 - 36 - 9 + 8 = -10$.

$\Rightarrow f_{min} = -10$

Q10 **If $x > 0$; $f(x) = 5x^2 + \frac{A}{x^5}$, $A > 0$ (constant), find smallest A such that $f(x) \geq 24 \forall x > 0$.**

Sol. Applying A.M $\geq$ G.M,

$$\frac{x^2 + x^2 + x^2 + x^2 + x^2 + \frac{A}{2x^5} + \frac{A}{2x^5}}{7} \geq \left(\frac{A^2}{4}\right)^{\frac{1}{7}}$$

$$f(x) \geq 7\left(\frac{A^2}{4}\right)^{\frac{1}{7}}$$

Also, $f_{min} \geq 24$

$$\Rightarrow 7\left(\frac{A^2}{4}\right)^{\frac{1}{7}} \geq 24 \Rightarrow \frac{A^2}{4} \geq \left(\frac{24}{7}\right)^7$$

$$\Rightarrow A^2 \geq 4\left(\frac{24}{7}\right)^7 \Rightarrow A \geq 2\left(\frac{24}{7}\right)^{\frac{7}{2}}$$

$$\Rightarrow A \in \left[2\left(\frac{24}{7}\right)^{\frac{7}{2}}, \infty\right)$$

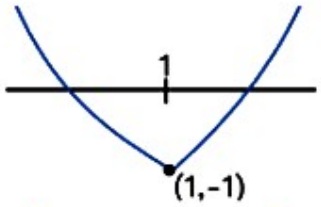
$\therefore$, smallest A possible is $2\left(\frac{24}{7}\right)^{\frac{7}{2}}$.

Q11 **Let $f(x) = \begin{cases} -x^3 + \frac{b^3 - b^2 + b - 1}{b^2 + 3b + 2}, & 0 \leq x < 1 \\ 2x - 3, & 1 \leq x \leq 3 \end{cases}$**

Find all possible real values of b such that f(x) has the smallest possible value at x = 1.

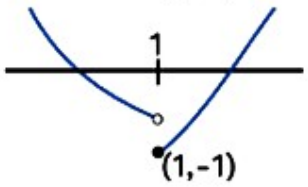
Sol. $f'(x) = \begin{cases} -3x^2, & 0 \leq x < 1 \\ 2, & 1 \leq x \leq 3 \end{cases}$

$\Rightarrow f(x)$ decreases in (0, 1) and then increases

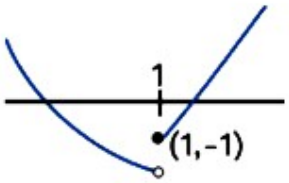


$\Rightarrow$ There are 3 possible cases depending on continuity of function:

Case-I : $\lim\limits_{x\to 1} f(x) = f(1)$



Case-II : $\lim\limits_{x\to 1} f(x) > f(1)$



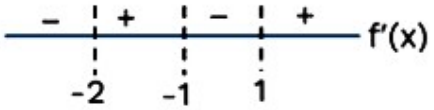
Case-III : $\lim\limits_{x\to 1} f(x) < f(1)$

It is clear that only case-I and case-II satisfies the condition.

$\Rightarrow \lim\limits_{x\to 1} f(x) \geq f(1) = -1$

$\Rightarrow -1 + \dfrac{b^3 - b^2 + b - 1}{b^2 + 3b + 2} \geq -1$

$\Rightarrow \dfrac{b^3 - b^2 + b - 1}{b^2 + 3b + 2} \geq 0$



$\Rightarrow \dfrac{(b^2+1)(b-1)}{(b+1)(b+2)} \geq 0$

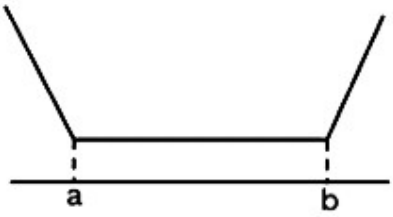
$\Rightarrow b \in (-2,-1) \cup [1,\infty)$

Q12 **$a, b \in R$ satisfying $|a-1|+|b-1| = |a|+|b| = |a+1| = |b+1|$. Find minimum value of $|a-b|, (a<b)$.**

Sol. Let $f(x) = |x-a|+|x-b|$

$\Rightarrow f(-1) = f(1) = f(0)$ (given)

$$\Rightarrow f(x) = \begin{cases} 2x-a-b, & x \geq b \\ b-a, & a \leq x < b \\ a+b-2x, & x < a \end{cases}$$



Since f(0) = f(−1) = f(1), it is only possible if −1, 0, 1 lies in [a, b]

Now $1-(-1) = 2$

$\therefore b - a \geq 2$

$\Rightarrow |a-b|_{min} = 2$ (when $a = -1$ and $b = 1$)

Q13 **Find all possible values of 'a' for which $f(x)=\log_a\left(4ax-x^2\right)$ is monotonically increasing for every $x\in\left[\frac{3}{2},2\right]$**

Sol. $f(x)=\dfrac{\ell n\left(4ax-x^2\right)}{\ell n(a)} \Rightarrow f'(x)=\dfrac{(4a-2x)}{\ell n(a)\left(4ax-x^2\right)}$

Now $\left(4ax-x^2\right)>0$ (Domain).

Case-I : $a>1\Rightarrow \ell n(a)>0$

$f'(x)\geq 0 \Rightarrow 4a-2x\geq 0$

$\Rightarrow x\leq 2a \Rightarrow x<2 \quad (a>1)$

$\Rightarrow$ condition is satisfied for $a>1$ ...(i)

Case-II : $a\in(0,1)\Rightarrow \ell n(a)<0$

$\Rightarrow f'(x)\geq 0\Rightarrow 4a-2x\leq 0$

$\Rightarrow x\geq 2a \Rightarrow a\leq \dfrac{x}{2}$

Since $x\in\left[\frac{3}{2},2\right]$, $a\leq\frac{3}{4}$

$\Rightarrow a\in\left(0,\frac{3}{4}\right]$ ...(ii)

Also, $4ax-x^2>0$ (domain)

$\Rightarrow 4a>x$

$\Rightarrow a>\dfrac{x}{4}$

$\Rightarrow a>\dfrac{\text{maximum possible values of } x}{4}$

$\Rightarrow a>\dfrac{2}{4}$

$\Rightarrow a>\dfrac{1}{2}$ ...(iii)

$(i)\cup(ii)\cup(iii)$

$\Rightarrow a\in\left(\frac{1}{2},\frac{3}{4}\right]\cup(1,\infty)$

ESTABLISHING INEQUALITIES :

Notion of monotonicity helps in establishing variety of inequalities involving algebraic and transcendental function with much greater ease.

If $f(x) \geq g(x)$ or $f(x) \leq g(x)$ is to be shown in some interval we create a new function $h(x) = f(x) - g(x)$ and using monotonicity check whether $h(x) \geq 0$ or $h(x) \leq 0$ in the given interval.

ILLUSTRATION:

Q1 **Prove that $2\sin x + \tan x \geq 3x \left(0 \leq x < \frac{\pi}{2}\right)$**

Sol. $f(x) = 2\sin x + \tan x - 3x$

$f'(x) = 2\cos x + \sec^2 x - 3$

$f''(x) = -2\sin x + 2\sec^2 x \tan x = 2\sin x[\sec^3 x - 1] \geq 0 \ \forall x \in \left[0, \frac{\pi}{2}\right)$

$\Rightarrow f'(x) \uparrow \text{ in} \left[0, \frac{\pi}{2}\right) \Rightarrow f'(x) \geq f'(0)$

or $f'(x) \geq 0 \Rightarrow f(x) \uparrow$ hence $f(x) \geq f(0)$

in $\left[0, \frac{\pi}{2}\right) \Rightarrow f(x) \geq 0 \qquad \Rightarrow 2\sin x + \tan x \geq 3x$

Q2 **Find the set of values of x for which $\ln(1 + x) > \frac{x}{1+x}$.**

Sol. $f(x) = \ln(1 + x) - \frac{x}{1+x} = \ln(1 + x) + \frac{1}{1+x} - 1$

Domain: $x > -1$

$f'(x) = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2}$

$f'(x) \geq 0 \ \forall \ x \geq 0 \Rightarrow f(x) \uparrow \ \& \ f'(x) \leq 0 \ \forall x \leq 0 \Rightarrow f(x) \downarrow$

$f'(0) = 0$

$\therefore f(x) > f(0) \ \forall x \in D_f - \{0\}$

$\therefore f(x) > 0 \ \forall \ x \in (-1, 0) \cup (0, \infty)$

Q3 **Show that $\ln(1+x) > x - \frac{x^2}{2}\ \forall x \in (0, \infty)$**

Sol. Consider the function $f(x) = \ln(1+x) - x + \frac{x^2}{2},\ x \in (0, \infty)$

$$f'(x) = \frac{1}{1+x} - 1 + \frac{1}{2}(2x)$$

$$= \frac{x^2}{x+1} > 0 \ \forall\ x > 0$$

$\Rightarrow$ f(x) strictly increases in $(0, \infty)$

$\Rightarrow$ $f(x) > f(0^+) = 0$

i.e. $\ln(1+x) > x - \frac{x^2}{2}$ which is the desired result.

Q4 **Show that the equation $x^5 - 3x - 1 = 0$ has a unique root in [1, 2].**

Sol. Consider the function

$f(x) = x^5 - 3x - 1,\ x \in [1, 2]$

and $f'(x) = 5x^4 - 3 > 0\ \forall\ x \in [1, 2]$

$\Rightarrow$ f(x) strictly increases in [1, 2]

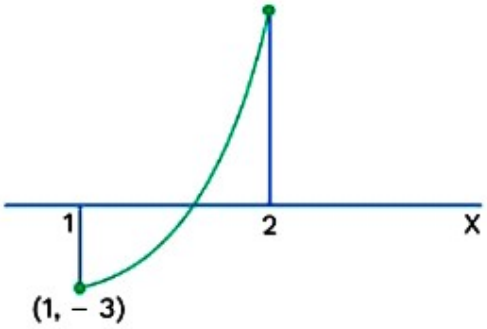
Also, we have

$f(1) = 1 - 3 - 1 = -3$

and $f(2) = 32 - 6 - 1 = 25$

From the shape of the curve shown alongside, we can see that the curve y = f(x) will cut the X-axis exactly once in [1, 2]

i.e. f(x) will vanish exactly once in [1, 2]



Q5 **Prove that $\frac{x}{1+x} < \ln(1+x) < x\ \forall\ x > 0$**

Sol. Consider the function $f(x) = \ln(1+x) - \frac{x}{1+x},\ x > 0.$

Then $f'(x) = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2} > 0\ \forall\ x > 0$

$\Rightarrow$ f(x) strictly increases in $(0, \infty)$

$\Rightarrow$ $f(x) > 0$, $f(0^+) = 0$ & i.e. $\ln(1+x) > \frac{x}{1+x}$ which proves the LHI.

Now, consider the function $g(x) = x - \ln(1+x),\ x > 0$

$$g'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x}$$

$\therefore g'(x) > 0\ \forall\ x > 0$

$\Rightarrow$ g(x) strictly increases in $(0, \infty) \Rightarrow g(x) > g(0^+) = 0$

i.e. $x > \ln(1+x)$ which proves the RHI.

Q6 **Using calculus, find the order of relation between x and $\tan^{-1}x$ when $x \in [0, \infty)$.**

Sol. Let $f(x) = x - \tan^{-1}x$

$\Rightarrow f(0) = 0;\ \ f'(x) = 1 - \frac{1}{1+x^2} = \frac{x^2}{1+x^2} \geq 0$

$\Rightarrow f(x)$ is increasing $\forall x \in [0, \infty)$

Since $f(0) = 0, f(x) > 0\ \forall\ x > 0$.

$\Rightarrow x - \tan^{-1}x \geq 0\ \ \forall\ x \geq 0$

$\Rightarrow x \geq \tan^{-1}x\ \ \forall\ x \geq 0$

Q7 **For $0 \leq p \leq 1$ and for any $a > 0, b > 0$, prove that $(a+b)^p \leq a^p + b^p$**

Sol. To prove: $(a+b)^p \leq a^p + b^p$

Or

$\left(\frac{a}{b} + 1\right)^p \leq \left(\frac{a}{b}\right)^p + 1$

Let $f(x) = (x+1)^p - x^p - 1,\ \ 0 \leq p \leq 1$

$\Rightarrow f'(x) = p(x+1)^{p-1} - px^{p-1}$

$= \frac{p}{(1+x)^{1-p}} - \frac{p}{x^{1-p}} \leq 0 \forall x > 0$

$\Rightarrow$ Since $f(0) = 0, f(x) \leq 0 \forall x > 0$

$\Rightarrow (x+1)^p - x^p - 1 \leq 0 \forall x > 0$

$\Rightarrow (x+1)^p \leq x^p + 1\ \forall\ x > 0$

$\Rightarrow \left(\frac{a}{b} + 1\right)^p \leq \left(\frac{a}{b}\right)^p + 1\ \forall a, b > 0$

$\Rightarrow (a+b)^p \leq a^p + b^p\ \ \forall a, b > 0, 0 \leq p \leq 1$ **Hence proved.**

Q8 **Proved that : $\frac{\tan x}{x} > \frac{x}{\sin x}$ for $0 < x < \frac{\pi}{2}$**

Sol. To proved : $\frac{\tan x}{x} > \frac{x}{\sin x}$ for $0 < x < \frac{\pi}{2}$

Or

$\sin x \tan x - x^2 > 0$ for $0 < x < \frac{\pi}{2}$

Let $f(x) = \sin x \tan x - x^2$

$f'(x) = \sin x \sec^2 x + \cos x \tan x - 2x$

$= \sin x (1 + \sec^2 x) - 2x$

$f''(x) = \cos x (1 + \sec^2 x) + \sin x (2 \sec x) \sec x \tan x - 2$

$= (\sec x + \cos x - 2) + \underbrace{2 \sec^2 x \sin x \tan x}_{>0}$
>0

$\Rightarrow f''(x) > 0 \ \forall x \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow f'(x)$ is increasing in $\left(0, \frac{\pi}{2}\right)$

Since $f'(x) = 0$ at $x = 0$, $f'(x) > 0$ in $x \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow f(x)$ is increasing in $\left(0, \frac{\pi}{2}\right)$

Again, since $f(0^+) = 0, f(x) > 0$ in $\left(0, \frac{\pi}{2}\right)$

$\Rightarrow \sin x \tan x - x^2 > 0$ in $\left(0, \frac{\pi}{2}\right)$

$\Rightarrow \frac{\tan x}{x} > \frac{x}{\sin x}$ in $\left(0, \frac{\pi}{2}\right)$ **Hence proved.**

Q9 **Establish the inequality given below by examining the sign of the derivative of an appropriate function: $\frac{1}{x + \left(\frac{1}{2}\right)} < \ell n \left(1 + \frac{1}{x}\right) < \frac{1}{x}$ for $x > 0$**

Sol. Let $f(x) = \frac{1}{x + \frac{1}{2}} - \ell n \left(1 + \frac{1}{x}\right)$

$$f'(x) = \frac{-1}{\left(x+\frac{1}{2}\right)^2} - \frac{1}{\left(1+\frac{1}{x}\right)}\left(\frac{-1}{x^2}\right)$$

$$= \frac{-4}{(2x+1)^2} + \frac{1}{x(x+1)}$$

$$= \frac{-4x^2 - 4x + 4x^2 + 4x + 1}{(2x+1)^2\ x(x+1)}$$

$$\Rightarrow f'(x) = \frac{1}{(2x+1)^2\, x(x+1)} > 0$$

$\Rightarrow f(x)$ is an increasing function

$f(\infty) = 0 - 0 = 0$

Since $f(\infty) \approx 0$ and f(x) is increasing function,

$\Rightarrow f(x) < 0 \forall x > 0$

$$\Rightarrow \frac{1}{x+\frac{1}{2}} - \ell n\left(1+\frac{1}{x}\right) < 0$$

$$\Rightarrow \frac{1}{x+\frac{1}{2}} < \ell n\left(1+\frac{1}{x}\right) \qquad \text{for } x > 0$$

Now Let $g(x) = \ell n\left(1+\frac{1}{x}\right) - \frac{1}{x}$

$$\Rightarrow g'(x) = \frac{-1}{x(x+1)} + \frac{1}{x^2}$$

$$= \frac{1}{x^2(x+1)} > 0 \forall x > 0$$

$\Rightarrow g(x)$ is increasing $\forall x > 0$

Again, $g(\infty) \approx 0$

Since g(x) is increasing $\forall x > 0$,

$g(x) < 0 \ \forall x > 0$

$$\Rightarrow \ell n\left(1+\frac{1}{x}\right) < \frac{1}{x} \forall x > 0$$

Hence proved.

Q10 **If $f(x)=\ell n(\ell n(x))$ where x > e, prove that**

$$\frac{1}{(m+1)\ell n(m+1)} \le f(m+1)-f(m) \le \frac{1}{m.\ell n(m)} \text{ for } m>e$$

Sol.

$$f'(x)=\frac{1}{\ell n(x)\,x}>0 \forall x>e$$

$$f''(x)=\frac{-1}{(x\ell n(x))^2}(\ell n(x)+1)<0 \forall x>e$$

$\Rightarrow f'(x)$ is a decreasing function

We know that $g(x)_{min}(b-a) \le \int_a^b g(x)dx \le g_{max}(b-a)$

Applying it on $f'(x)$ in $(m,m+1)$

$$f'_{min}(m+1-m) \le \int_m^{m+1} f'(x)dx \le f'_{max}(m+1-m)$$

Since $f'(x)$ is decreasing function, $f'(x)_{min}=\frac{1}{(m+1)\ell n(m+1)}$ and $f'_{max}=\frac{1}{m\ell n(m)}$

$$\Rightarrow \frac{1}{(m+1)\ell n(m+1)} \le f(m+1)-f(m) \le \frac{1}{m\ell n(m)} \forall m>e$$ **Hence proved.**

Q11 **Prove that $f(x)=\left(1+\frac{1}{x}\right)^x$ is always an increasing function for all x in its domain. State its range and also plot the graph of the function.**

Sol. Domain of $f^x \to f>0$

$$\therefore 1+\frac{1}{x}>0$$

$$\Rightarrow \frac{1+x}{x}>0$$

$$\Rightarrow x \in (-\infty,-1)\cup(1,\infty)$$

$$\lim_{x\to\infty} f(x)=e^{\lim_{x\to\infty} x\left(1+\frac{1}{x}-1\right)}$$

$$\Rightarrow f(-\infty)=e$$

$$f(-1^-)=\infty$$

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} e^{x\ell n\left(1+\frac{1}{x}\right)}$$

$$= \lim_{x\to\infty} e^{\left(\frac{\ell n\left(1+\frac{1}{x}\right)}{\frac{1}{x}}\right)}$$

$$\Rightarrow f(\infty) = e$$

$$\Rightarrow \lim_{x\to\infty} f(x) = e$$

$$f(x) = e^{x\ell n\left(1+\frac{1}{x}\right)}$$

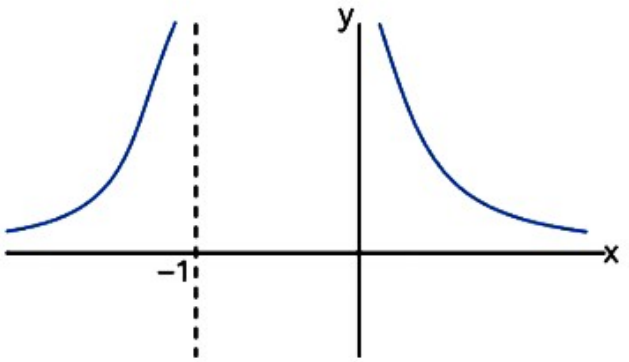
$$\Rightarrow f'(x) = e^{x\ell n\left(1+\frac{1}{x}\right)}\left[\ell n\left(1+\frac{1}{x}\right)+\left(\frac{x}{\left(1+\frac{1}{x}\right)}\right)\left(\frac{-1}{x^2}\right)\right]$$

$$\Rightarrow f'(x) = \left(1+\frac{1}{x}\right)^x\left(\ell n\left(1+\frac{1}{x}\right)-\frac{1}{x+1}\right)$$

Let $g(x) = \ell n\left(1+\frac{1}{x}\right)-\frac{1}{x+1}$

$$g'(x) = \left(\frac{1}{1+\frac{1}{x}}\right)\left(\frac{-1}{x^2}\right)+\frac{1}{(x+1)^2}$$

$$= \frac{-1}{x(x+1)}+\frac{1}{(x+1)^2}$$



$\Rightarrow g(x)$ will decrease for $x > 0$ and

increase for $x < 0$

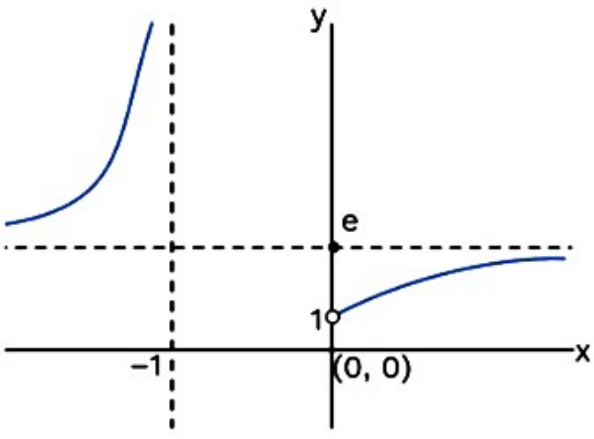
Since, g(x) increases for $x < 0$ and then decreases for $x > 0$

We can conclude that $g(x) > 0$

$\Rightarrow f'(x) > 0 \Rightarrow f(x)$ is increasing.

$\Rightarrow$ Graph of f(x)

Range : $(1, \infty) - \{e\}$



Q12 **Prove that sin1 > cos(sin1). Also show that $\sin\left(\cos\left(\sin x\right)\right)=\cos\left(\sin\left(\cos x\right)\right)$ has only one solution in $\left[0,\frac{\pi}{2}\right]$**

Sol. $\sin 1 \approx \sin 57^\circ \approx \frac{\sqrt{3}}{2} \approx 0.85$

$\frac{\pi}{2}-1 \approx 0.57$

Since cos x is decreasing in $\left(0,\frac{\pi}{2}\right)$,

$\cos\left(\frac{\pi}{2}-1\right) > \cos\left(\sin 1\right)$

$\Rightarrow \sin 1 > \cos\left(\sin 1\right)$ ...(i)

Let $g\left(x\right) = \sin\left(\cos\left(\sin x\right)\right) - \cos\left(\sin\left(\cos x\right)\right)$

$g\left(0\right) = \sin 1 - \cos\left(\sin 1\right) > 0$ (from (i))

$g\left(\frac{\pi}{2}\right) = \sin\left(\cos 1\right) - 1 < 0$

$g'\left(x\right) = \cos\left(\cos\left(\sin x\right)\right)\left(-\sin\left(\sin x\right)\right)\cos x + \sin\left(\sin\left(\cos x\right)\right)\cos\left(\cos x\right)\left(-\sin x\right)$

$\Rightarrow g'\left(x\right) < 0$ in $x \in \left(0,\frac{\pi}{2}\right)$

$\Rightarrow$ g(x) is monotonically decreasing in $\left[0,\frac{\pi}{2}\right]$

Using above information with $g\left(0\right) > 0$ and $g\left(\frac{\pi}{2}\right) < 0$,

We can say that g(x) will only have one root in $\left[0,\frac{\pi}{2}\right]$ **Hence proved.**

Rolle's Theorem :

Let f(x) be a function subjected to the following conditions:

(i) f(x) is a continuous function in the closed interval of $a \le x \le b$.

(ii) f'(x) exists for every point in the open interval $a < c < b$

(iii) f(a) = f(b)

then there is some point $c \in (a, b)$ such that f'(c) = 0.

Alternative statement :

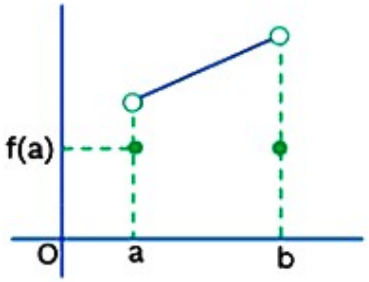
Rolles theorem states that between any two real zeroes of a differentiable real function f, lies at least one critical point of f'(x).

Remarks :

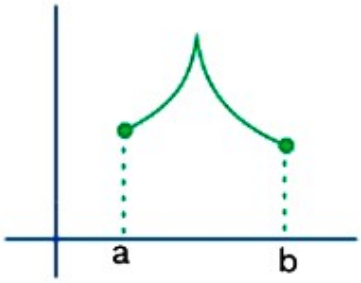
(i) Converse of Rolle's theorem is Not true i.e. f'(x) may vanish at a point within (a, b) without satisfying all the three condition of Rolle's Theorem.

(iii) The three conditions are sufficient but not necessary for f'(x) = 0 for some x in (a, b)

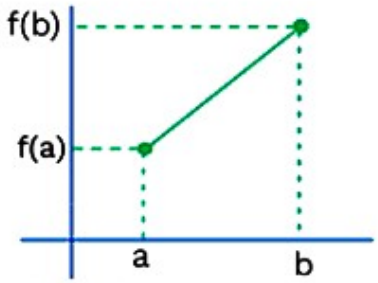
(iii) If the function y = f(x) defined over [a, b] does not satisfy even one of the 3 conditions then Rolle's Theorem fails i.e. there may or may not exist point in (a, b) where f'(x) = 0.



f is not continuous in [a, b] then Rolles theorem fails



f is not derivable in open (a, b)



$f(a) \neq f(b)$

Ex **Check if Rolle's theorem is possible in following functions. Also find 'C' if possible.**

(i) $f(x) = x^2$ in $[-1,1]$ **(ii)** $f(x) = x^3 - 3x^2 + 2x + 5$ in $[0,2]$

Sol. **(i)** f(x) is continuous and derivable in [–1, 1]

$f(-1) = 1 = f(1)$

$\Rightarrow$ Rolle's theorem is possible for f(x).

$\Rightarrow f'(x) = 0$ for some 'C'

$\Rightarrow 2x = 0$ for some C

$\Rightarrow$ C = 0.

(ii) f(x) is continuous and derivable as it is a polynomial.

$f(0) = 5 = f(2)$

$\Rightarrow$ Rolle's theorem is applicable

$\Rightarrow f'(c) = 0$

$\Rightarrow 3c^2 - 6c + 2 = 0$

$\Rightarrow c = \frac{6 \pm \sqrt{12}}{6}$

$\Rightarrow c = 1 \pm \frac{1}{\sqrt{3}}$

Since both values lie in $[0,2]$, $c = 1 \pm \frac{1}{\sqrt{3}}$

LMVT THEOREM
(LAGRANGE'S MEAN VALUE THEOREM)

Let f(x) be a function of x subject to the following conditions:

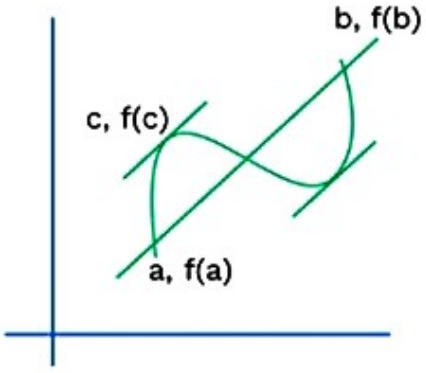
(i) f(x) is a continuous function of x in the closed interval of $a \le x \le b$.

(ii) f'(x) exists for every point in the open interval $a < x < b$.

(iii) $f(a) \neq f(b)$.

Then there exists at least one point x = c such that $a < c < b$ where $f'(c) = \frac{f(b) - f(a)}{b - a}$

Geometrically, the slope of the secant line joining the curve at x = a & x = b is equal to the slope of the tangent line drawn to the curve at x = c. Note the following:



Note :

Now [f(b) – f(a)] is the change in the value of function f as x changes from a to b so that [f(b) – f(a)]/ (b – a) is the average rate of change of the function at x = c. Thus, the theorem states that the average rate of change of a function over an interval is also the actual rate of change of the function at some point

of the interval. In particular, for instance, the average velocity of a particle over an interval of time is equal to the velocity at some instant belonging to the interval.

This interpretation of the theorem justifies the name "Mean Value" for the theorem.

Rolles theorem is a special case of LMVT since

$f(a) = f(b) \Rightarrow f'(c) = \dfrac{f(b) - f(a)}{b - a} = 0.$

Alternative form of LMVT :

Another form of statement of Lagrange's Mean Value Theorem. If a function f is continuous in a closed interval [a, a + h] and derivable in the open interval (a, a + h), then there exists at least one number 'θ' $\in$ (0, 1) such that
$f(a + h) = f(a) + hf'(a + \theta h)$. $\theta \in (0, 1)$

Proof :

We write b – a = h so that h denotes the length of the interval [a, b] which may now be rewritten as [a, a + h]. The number, 'c' which lies between a and a + h, is greater than a and less than a + h so that we may write $c=a+\theta h$, where θ is some number between 0 and 1.

Thus the equation (i) becomes $\dfrac{f(a+h) - f(a)}{h}$

$= f'(a + \theta h) \Rightarrow f(a + h) = f(a) + hf'(a + \theta h)$

ILLUSTRATION :

Q. **Verify Rolle's Theorem for**

(a) $f(x) = x(x + 3)e^{-x/2}$ **in** $[-3, 0]$

(b) $f(x) = f(x) = \dfrac{\sin x}{e^x}$ **in** $[0, \pi]$

(c) $f(x) = 1 - x^{2/3}$ **in** $[-1, 1]$ **(f'(0) non existent)**
Also find 'c' if possible

Sol. **(a)** $f(x) = x(x+3)e^{-x/2}, \quad x \in [-3, 0]$

$f(-3) = 0 \; ; \quad f(0) = 0$

Clearly f(x) is continuous and differentiable function

So, there exists

$c \in (-3, 0)$ s. t. $f'(c) = 0$

$$f'(c) = (2c+3)e^{-c/2} - \frac{1}{2}(c^2+3c)e^{-c/2}$$

$$\Rightarrow e^{-c/2}\left[2c+3-\frac{(c^2+3c)}{2}\right] = 0$$

$4c + 6 - c^2 - 3c = 0$

$c^2 - c - 6 = 0$

$c = -2; \; c = 3 \notin (-3, 0)$

(b) $f(x) = \dfrac{\sin x}{e^x}, \quad x \in [0, \pi]$

$f(0) = f(\pi) = 0$

Continuous and differentiable function.

$f'(c) = 0 \Rightarrow e^c \cos c - \sin c \, e^c = 0$

$\tan c = 1 \Rightarrow c = \dfrac{\pi}{4} \in (0, \pi)$

(c) $f(x) = 1 - x^{2/3}$ in $[-1, 1]$

$f(-1) = f(1) = 0$

$f'(c) = \dfrac{-2}{3}c^{-1/3}$ but for $c = 0$ $f'(c)$ is non-existent.

$\therefore$ f(x) is not differentiable in given interval.

Hence Rolle's theorem is not applicable.

Q. **Find c of LMVT for $f(x) = \sqrt{x-1}$ in [1, 3]**

Sol. $f(x) = \sqrt{x-1}$ in $[1, 3]$

Continuous and diff. in given interval

$$f'(c) = \frac{f(3) - f(1)}{3-1}$$

$$\frac{1}{2\sqrt{c-1}} = \frac{\sqrt{2}}{2} \Rightarrow c = \frac{3}{2}$$

Q. **Using LMVT prove that $|\cos a - \cos b| \le |a - b|$**

Sol. Consider f(x) = cosx in [a, b]. Applying LMVT,

$$\left|\frac{\cos b - \cos a}{b - a}\right| = |-\sin c| \le 1 \Rightarrow |\cos b - \cos a| \le |b - a|$$

$|\cos a - \cos b| \le |a - b|$ **Hence Proved.**

Q1 **The value of c of Rolle's theorem for the function $f(x) = (x-a)^m (x-b)^n, m, n \in N, x \in [a, b]$ is**

(A) $\frac{ma + nb}{m + n}$ **(B)** $\frac{mb + na}{m + n}$ **(C)** $\frac{mb - na}{m - n}$ **(D) None of these**

Sol. **(B)**

f(x) is always continuous and derivable.

Also, f(a) = f(b) = 0

$\Rightarrow$ Rolle's theorem is applicable.

$\Rightarrow f'(c) = 0$ for some c in (a, b)

$$\Rightarrow m(x-a)^{m-1}(x-b)^n + n(x-a)^m (x-b)^{n-1} = 0$$

$$\Rightarrow m(x-b) + n(x-a) = 0$$

$$\Rightarrow (m+n)x = mb + na$$

$$\Rightarrow x = \frac{mb + na}{m + n} = c.$$

Q2 **If the function $f(x) = x^3 - 6x^2 + ax + b$ defined on $[1, 3]$ satisfies the Rolle's theorem for $c = \frac{2\sqrt{3} + 1}{\sqrt{3}}$, then find the value of a and b.**

Sol. $f(1) = f(3)$

$$\Rightarrow a + b - 5 = 3a + b - 27$$

$$\Rightarrow 2a = 22$$

$$\Rightarrow a = 11$$

$$f'(c) = 0$$

$$f'(x) = 3x^2 - 12x + a$$

$$= 3x^2 - 12x + 11$$

It has one root $c = 2 + \frac{1}{\sqrt{3}}$

$\Rightarrow$ Other root must be $2 - \frac{1}{\sqrt{3}}$

$\Rightarrow$ Sum of the roots $= 4 = \frac{12}{3}$ and product $= \frac{11}{3}$

Hence, it is always true

$\Rightarrow a = 11,\ b \in R$.

Q3 **If $f(x) = x^{\alpha} \ln(x)$ and f(0) = 0, then value of α for which Rolle's theorem can be applied in [0, 1] is**

(A) −2 **(B) −1** **(C) 0** **(D) $\frac{1}{2}$**

Sol. **(D)**

f(x) is continuous in [0, 1] and derivable in (0, 1).

$\lim\limits_{x \to 0^+} f(x) = f(0) = 0$

$\Rightarrow \lim\limits_{x \to 0} x^{\alpha} \ln(x) = 0$

$\Rightarrow \lim\limits_{x \to 0} \frac{\ell n(x)}{x^{-\alpha}} = 0$

$\Rightarrow$ It is true for $\alpha = \frac{1}{2}$ only.

Q4 **Consider the function $f(x) = \begin{cases} x \sin\frac{\pi}{x}, & x > 0 \\ 0, & x = 0 \end{cases}$ then the number of points in (0, 1), where the derivative is zero is**

(A) 0 **(B) 1** **(C) 2** **(D) infinite**

Sol. **(D)**

f(x) is continuous and derivable for x > 0

$f\left(\frac{1}{2}\right) = f\left(\frac{1}{3}\right) = f\left(\frac{1}{4}\right) = \ldots f\left(\frac{1}{n}\right) = 0, n \in N$

$\Rightarrow f(x) = 0$ have infinite roots in (0, 1).

So, in between every of these consecutive roots, Rolle's theorem can be applied.

For ex. Rolle's can be applied in $\left(\frac{1}{3}, \frac{1}{2}\right)$ or $\left(\frac{1}{4}, \frac{1}{3}\right)$ or $\left(\frac{1}{5}, \frac{1}{4}\right)$ etc.

$\Rightarrow$ There are infinite intervals with each interval having a point where $f'(x) = 0$

$\Rightarrow$ $f'(x) = 0$ will have infinite roots in (0, 1)

Q5 **$f(x) = 4x^3 - 3x^2 - 2x + 1$. Use Rolle's theorem to prove that for atleast one $c \in (0,1), f(c) = 0$**

Sol. f(x) is always continuous and derivable.

Let $g(x) = \int f(x)\,dx = x^4 - x^3 - x^2 + x + k$

g(x) is also continuous and derivable function

Also, g(0) = k = g(1)

$\therefore$, By Rolle's theorem, $g'(x) = 0$ has atleast one root in (0, 1).

Since $g'(x) = f(x)$, it implies that $f(x) = 0$ has alteast one root in (0, 1)

Hence proved.

Q6 **Given $a_0, a_1, a_2 \ldots a_n$ are real numbers satisfying the equation $\frac{a_0}{n+1} + \frac{a_1}{n} + \frac{a_2}{n-1} + \ldots + \frac{a_{n-1}}{2} + a_n = 0$ then prove that there exists at least one real x between 0 and 1 such that $a_0x^n + a_1x^{n-1} + \ldots + a_n = 0$**

Sol. Let $f(x) = \frac{a_0x^{n+1}}{n+1} + \frac{a_1x^n}{n} + \frac{a_2x^{n-1}}{n-1} + \ldots a_nx$

$\Rightarrow$ $f(x)$ is continuous and derivable

Also f(0) = 0 = f(1) (given).

$\Rightarrow$ Rolle's theorem is applicable

$\Rightarrow$ $f'(x) = 0$ has one root in (0, 1)

$\Rightarrow$ $a_0x^n + a_1x^{n-1} + \ldots + a_n = 0$ has one root in (0, 1). **Hence proved.**

Q7 **Let $f(x) = xe^x - 1$. Use Rolle's theorem to prove that the equation f(x) = 0 has one and only one root in (0, 1).**

Sol. Let $g(x) = \int f(x)\,dx$

$= \int (xe^x - 1)\,dx$

$= xe^x - e^x - x + c$

Now, g(x) is both continuous and derivable.

$g(0) = 0 - 1 - 0 + c = c - 1 = g(1)$

$\therefore$ we can apply Rolle's theorem on $g(x)$ in (0, 1)

$\Rightarrow g'(x) = 0$ will have atleast one root in (0, 1)

Since $g'(x) = f(x) \Rightarrow f(x)$ has atleast one root in (0, 1).

Alternatively

$f'(x) = xe^x + e^x = e^x(x+1) > 0 \ \forall x \in (0,1)$.

$\Rightarrow f(x)$ is monotonic increasing on (0, 1)

$\Rightarrow f(x)$ can't have more than one root

So, f(x) will only have one root in (0, 1).

Note:

If a function is equal at 6 distinct points, (function is continuous and derivable), then

$\rightarrow f'(x) = 0$ has atleast 5 solution.

$\rightarrow f''(x) = 0$ has atleast 4 solution.

$\rightarrow f'''(x) = 0$ has atleast 3 solution.

Q8 **Suppose that the second derivative of f exists everywhere and that $f(x_1) = f(x_2) = f(x_3) = 0$ where $x_1 < x_2 < x_3$. Show that $f''(c) = 0$ for some number c with $x_1 < c < x_3$.**

Sol. Since f(x) is continuous and derivable and $f(x_1) = f(x_2) = 0$, by applying Rolle's theorem,

$\Rightarrow f'(c_1) = 0$ where $x_1 < c_1 < x_2$

Similarly since $f(x_2) = f(x_3) = 0$, there exists $c_2 \in (x_2, x_3)$ such that $f'(c_2) = 0$.

Now, $f'(c_1) = f'(c_2) = 0$

Again applying Rolle's theorem on $f'(x)$ in (c_1, c_2)

$\Rightarrow f''(c) = 0$ for some c in (c_1, c_2)

Since c_1 and c_2 lies in (x_1, x_3), we can say that $f''(c) = 0$ for some 'c' in (x_1, x_3). **Hence proved.**

Q9 **Let P(x) be a polynomial with real coefficients. Let $a, b \in R, a < b$, be two consecutive roots of P(x). Show that there exists 'c' such that $a \le c \le b$ and $P'(c) + 100P(c) = 0$**

Sol. Let $g(x) = P(x)e^{100x}$

$\Rightarrow g(a) = g(b) = 0 \qquad (P(a) = P(b) = 0;\ \text{given})$

So, applying Rolle's theorem on g(x) in (a, b),

$\Rightarrow g'(c) = 0$ for some 'c' in (a, b)

$\Rightarrow P'(c)e^{100c} + 100P(c)e^{100c} = 0$ for some 'c' in (a, b)

$\Rightarrow P'(c) + 100P(c) = 0$ for some 'c' in (a, b) **Hence proved.**

Q10 **Show that between any two roots of the equation $e^x \cos x = 1$, there exists atleast one root of $e^x \sin x - 1 = 0$.**

Sol. $e^x \cos x = 1 \Rightarrow \cos x = e^{-x}$

$\Rightarrow$ Let $f(x) = e^{-x} - \cos x$

$\Rightarrow f(x)$ have two roots, say 'α' and 'β'.

Applying Rolle's on f(x) in (α, β),

$\Rightarrow f'(c) = 0$ for some 'c' in (α, β)

$\Rightarrow -e^{-c} + \sin c = 0 \ , c \in (\alpha, \beta)$

$\Rightarrow e^c \sin c - 1 = 0$ for some c in (α, β) **Hence proved.**

Q11 $f:\left[0,\frac{\pi}{2}\right]\to(0,1)$ **is a differentiable function such that** $f(0)=0$, $f\left(\frac{\pi}{2}\right)=1$.

Then which of the following is correct?

(A) $f'(\alpha)=\frac{2}{\pi}$ **for atleast on** $\alpha\in\left(0,\frac{\pi}{2}\right)$

(B) $f(\alpha)f'(\alpha)=\frac{1}{\pi}$ **for atleast one** $\alpha\in\left(0,\frac{\pi}{2}\right)$

(C) $f'(\alpha)=\frac{8\alpha}{\pi^2}$ **for atleast one** $\alpha\in\left(0,\frac{\pi}{2}\right)$

(D) $f'(\alpha)=\sqrt{1-(f(\alpha))^2}$ **for atleast one** $\alpha\in\left(0,\frac{\pi}{2}\right)$

Sol. **(ABCD)**

(A) Let $g(x)=f(x)-\frac{2x}{\pi}$

$\Rightarrow g(0)=0=g\left(\frac{\pi}{2}\right)$ $\left(f\left(\frac{\pi}{2}\right)=1\right)$

$\therefore$ Applying Rolle's theorem on g(x),

$g'(x)=0$ for atleast one value in $\left[0,\frac{\pi}{2}\right]$

$\Rightarrow f'(c)-\frac{2}{\pi}=0$ for atleast one $c\in\left(0,\frac{\pi}{2}\right)$

$\therefore$ (A) is correct

(B) Let $g(x)=\frac{(f(x))^2}{2}-\frac{x}{\pi}$

$\Rightarrow g(0)=0=f\left(\frac{\pi}{2}\right)$.

$\Rightarrow$ Applying Rolle's on g(x) in $\left[0,\frac{\pi}{2}\right]$.

$g'(x)=0$ for atleast one α in $\left(0,\frac{\pi}{2}\right)$

$\Rightarrow f(\alpha)f'(\alpha)-\frac{1}{\pi}=0$ for atleast one α in $\left(0,\frac{\pi}{2}\right)$.

$\Rightarrow$ (B) is correct.

(C) Let $g(x) = f(x) - \frac{4x^2}{\pi^2}$

$\Rightarrow g(0) = 0 = g\left(\frac{\pi}{2}\right)$

$\Rightarrow$ Applying Rolle's theorem on g(x) in $\left[0, \frac{\pi}{2}\right]$,

$\Rightarrow g'(\alpha) = 0$ for some α in $\left(0, \frac{\pi}{2}\right)$.

$\Rightarrow$ (C) is correct option.

(D) $f'(x) = \sqrt{1-(f(x))^2}$

$\Rightarrow \frac{f'(x)}{\sqrt{1-(f(x))^2}} = 1$

Let $g(x) = \sin^{-1}(f(x)) - x$

$\Rightarrow g(0) = 0 = g\left(\frac{\pi}{2}\right)$

$\Rightarrow$ Applying Rolle's theorem on g(x) in $\left[0, \frac{\pi}{2}\right]$

$g'(\alpha) = 0$ for atleast one α in $\left(0, \frac{\pi}{2}\right)$

$\Rightarrow f'(\alpha) = \sqrt{1-(f(\alpha))^2}$ for atleast one α in $\left(0, \frac{\pi}{2}\right)$

$\Rightarrow$ (D) is correct option.

Q12 **Consider the function f(x) = 8x² − 7x + 5 on the interval [−6, 6]. Find the value of c that satisfies the conclusion of Lagrange's mean value theorem.**

Sol. f(x) is continuous and differentiable
Applying LMVT on f(x) in $(-6, 6)$,

$$f'(c) = 16c - 7 = \frac{f(6) - f(-6)}{6 - (-6)}$$

$$\Rightarrow 16c - 7 = \frac{-7(6 - (-6))}{12}$$

$\Rightarrow 16c - 7 = -7 \Rightarrow c = 0$

Q13 **If a < b, show that a real number 'c' can be found in (a, b) such that $3c^2 = a^2 + ab + b^2$**

Sol. $3c^2 = a^2 + ab + b^2 = \frac{b^3 - a^3}{b - a}$

Let $f(x) = x^3$ in (a, b).

Since f(x) is continuous and derivable, applying LMVT,

$\Rightarrow f'(c) = 3c^2 = \frac{b^3 - a^3}{b - a}$ for some 'c' in (a, b).

$\Rightarrow 3c^2 = a^2 + b^2 + ab$ **Hence proved.**

Q14 **Use LMVT to prove that $\tan x > x$ $x \in \left(0, \frac{\pi}{2}\right)$**

Sol. Let $f(x) = \tan x$

In interval $(0, x)$, $x < \frac{\pi}{2}$,

Applying LMVT, we get:

$\sec^2 c = \frac{\tan x - \tan 0}{x - 0}$ for some c in (0, x)

$\Rightarrow \frac{\tan x}{x} = \sec^2 c > 1$ for $x \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow \tan x > x$ for $x \in \left(0, \frac{\pi}{2}\right)$ **Hence proved.**

Q15 **Let f be differentiable for all x. If $f(1) = -2$ and $f'(x) \geq 2$ for $x \in [1, 6]$, then find the range of value of f(6).**

Sol. $f'(x) \geq 2$

$\Rightarrow \int_1^6 f'(x)\,dx \geq \int_1^6 2\,dx$

$\Rightarrow f(6) - f(1) \geq 2(6 - 1)$

$\Rightarrow f(6) + 2 \geq 10$

$\Rightarrow f(6) \geq 8$

$\Rightarrow f(6) \in [8, \infty)$

Q16 **f is differentiable in [0, 2], f(0) = 2, f(2) = 8, $f'(x) \le 3 \forall x \in (0,2)$. f(1) = ?**

Sol. $f'(x) \le 3$

$$\Rightarrow \int_0^1 f'(x)dx \le \int_0^1 3dx$$

$$\Rightarrow f(1) - f(0) \le 3$$

$$\Rightarrow f(1) \le 5 \quad \ldots(i)$$

$$\int_1^2 f'(x)\,dx \le \int_1^2 3dx$$

$$\Rightarrow f(2) - f(1) \le 3$$

$$\Rightarrow 8 - f(1) \le 3$$

$$\Rightarrow f(1) \ge 5 \quad \ldots(ii)$$

From (i) & (ii), we say that above conditions are only possible if f(1) = 5.

Q17 **Suppose f and g be two differentiable functions in [a, b] where $0 < a < b$. If $g(x) \neq 0$ for all x in [a, b], prove that there exists 'c' such that $a < c < b$ and $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$. Use this results to find whether 'c' is an A.M/G.M/H.M/ none between 'a' and 'b' for the pair of functions given below in [a, b].**

(i) $f(x) = \frac{1}{x^2}$ **and** $g(x) = \frac{1}{x}$

(ii) $f(x) = e^x$ **and** $g(x) = e^{-x}$

(iii) $f(x) = \sqrt{x}$ **and** $g(x) = \frac{1}{\sqrt{x}}$

Sol. **(i)** To prove : $\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$

Or

To prove: $f'(c)\big(g(b)-g(a)\big) - g'(c)\big(f(b)-f(a)\big) = 0$

Let $h(x) = f(x)\big(g(b)-g(a)\big) - g(x)\big(f(b)-f(a)\big)$

$h(a) = f(a)\big(g(b)-g(a)\big) - g(a)\big(f(b)-f(a)\big)$

$$h(b) = f(b)\big(g(b) - g(a)\big) - g(b)\big(f(b) - f(a)\big)$$

$$\Rightarrow h(a) = h(b) = f(a)g(b) - f(b)g(a).$$

$\therefore$, we can apply Rolle's theorem on h(x)

$$\Rightarrow h'(c) = 0 \text{ for some c in (a, b)}$$

$$\Rightarrow f'(c)\big(g(b) - g(a)\big) = g'(c)\big(f(b) - f(a)\big)$$

$$\Rightarrow \frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)} \text{ for some c in (a, b)}$$ **Hence proved.**

Now, we are given $f(x) = \frac{1}{x^2}$ and $g(x) = \frac{1}{x}$

$$\Rightarrow \frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

$$\Rightarrow \frac{-\frac{2}{x^3}}{\frac{-1}{x^2}} = \frac{\frac{1}{b^2} - \frac{1}{a^2}}{\frac{1}{b} - \frac{1}{a}} \quad \text{for x = c in (a, b)}$$

$$\Rightarrow \frac{2}{c} = \frac{1}{b} + \frac{1}{a} \quad \text{for some c in (a, b)}$$

$$\Rightarrow a, c, b \rightarrow \text{H.M.}$$

(ii) Again $\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$ for some c in (a, b)

$$\Rightarrow \frac{e^c}{-e^{-c}} = \frac{e^b - e^a}{e^{-b} - e^{-a}}$$

$$\Rightarrow -e^{2c} = -e^a e^b$$

$$\Rightarrow 2c = a + b$$

$$\Rightarrow a, c, b \rightarrow \text{A.M.}$$

(iii) Again $\frac{f'(c)}{g'(c)} = \frac{\frac{1}{2\sqrt{c}}}{\frac{-1}{2c\sqrt{c}}} = \frac{\sqrt{b} - \sqrt{a}}{\frac{1}{\sqrt{b}} - \frac{1}{\sqrt{a}}}$

$$\Rightarrow -c = -\sqrt{ab}$$

$$\Rightarrow c^2 = ab$$

$$\Rightarrow a, c, b \rightarrow \text{G.M.}$$

Q18 **If $y = f(x)$ is twice differentiable function on [a, b]such that f(a) = f(b) = 0 and $f(x) > 0\ \forall\ x \in (a,b)$, then**

(A) $f''(c) < 0$ for some $c \in (a,b)$ **(B) $f''(c) > 0\ \forall\ c \in (a,b)$**

(C) $f(c) = 0$ for some $c \in (a,b)$ **(D) None of these**

Sol. **(A)**

In above case two types of graph

are possible:

From above graph we can see that,

(A) is true in both case

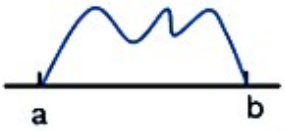
(B) is only true in (ii) case

(C) is not possible in both case

(i)



(ii)



Q19 **If f(x) is continuous in [a, b] and differentiable in (a, b), prove that there is atleast one $c \in (a,b)$ such that $\frac{f'(c)}{3c^2} = \frac{f(b) - f(a)}{b^3 - a^3}$**

Sol. Let $g(x) = x^3$ in [a, b].

$\Rightarrow g(x)$ is continuous and derivable.

$\Rightarrow \frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$ for some c in (a, b)

$\Rightarrow \frac{f'(c)}{3c^2} = \frac{f(b) - f(a)}{b^3 - a^3}$ for some c in (a, b) **Hence proved.**

Q20 **Let f(x) and g(x) be two differentiable functions in R and f(2) = 8, g(2) = 0, f(4) = 10 and g(4) = 8, then prove that $g'(x) = 4\ \ f'(x)$ for atleast one x ∈ (2,4).**

Sol. Again, using $\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$ for some c in (b, a),

$\Rightarrow \frac{f'(c)}{g'(c)} = \frac{f(4) - f(2)}{g(4) - g(2)}$ for some c in (2, 4)

$$\Rightarrow \frac{f'(c)}{g'(c)} = \frac{10-8}{8-0} = \frac{1}{4} \text{ for some c in (2, 4)}$$

$$\Rightarrow g'(x) = 4f'(x) \text{ for atleast one } x \in (2,4).$$ **Hence proved.**

Q21 **Let a, b, c be three real numbers such that $a < b < c$, $f(x)$ is continuous in [a, c] and differentiable in (a, c). Also $f'(x)$ is strictly increasing in (a, c). Prove that $(b-c)f(a)+(c-a)f(b)+(a-b)f(c) < 0$.**

Sol. Using LMVT, $f'(\lambda_1) = \frac{f(b)-f(a)}{b-a}, \lambda_1 \in (a,b)$

Similarly, $f'(\lambda_2) = \frac{f(c)-f(b)}{c-b}$ for same $\lambda_2 \in (b,c)$

Since $f'(x)$ is increasing and $\lambda_2 > \lambda_1$

$$\Rightarrow f'(\lambda_2) > f'(\lambda_1)$$

$$\Rightarrow \frac{f(c)-f(b)}{c-b} > \frac{f(b)-f(a)}{b-a}$$

$$\Rightarrow (b-a)f(c)-(b-a)f(b) > (c-b)f(b)-(c-b)f(a)$$

$$\Rightarrow (c-b)f(a)+(b-a)f(c)+(a-c)f(b) > 0$$

$$\Rightarrow (b-c)f(a)+(c-a)f(b)+(a-b)f(c) < 0$$ **Hence proved.**

Q22 **If the function $f:[0,4] \to R$ is differentiable then show that**

(i) for $(a,b) \in (0,4)$, $f^2(4)-f^2(0) = 8f'(a)f(b)$

(ii) $\int_0^4 f(t)dt = 2\left[\alpha f(\alpha^2)+\beta f(\beta^2)\right] \forall 0 < \alpha, \beta < 2$

Sol. (i) Using LMVT, $\frac{f(4)-f(0)}{4-0} = f'(a)$ for some $a \in (0,4)$

Also, using Intermediate value theorem,

$$\frac{f(4)+f(0)}{2} = f(b) \text{ for some } b \in (0,4)$$

$$\Rightarrow \left(\frac{f(4)-f(0)}{4}\right)\left(\frac{f(4)+f(0)}{2}\right) = f'(a)f(b)$$

$$\Rightarrow f^2(4)-f^2(0) = 8f'(a)f(b) \text{ for } a,b \in (0,4)$$ **Hence proved.**

(ii) Let $g(x)=\int_0^{x^2} f(t)\,dt$.

If $\alpha \in (0,1)$, then using LMVT,

$$g'(\alpha)=\frac{g(1)-g(0)}{1-0} \qquad \ldots(i)$$

If $\beta \in (1,2)$, then using LMVT,

$$g'(\beta)=\frac{g(2)-g(1)}{2-1} \qquad \ldots(ii)$$

(i) + (ii)

$$\Rightarrow g'(\alpha)+g'(\beta)=g(2)-g(0)$$

$$\Rightarrow 2\alpha f(\alpha^2)+2\beta f(\beta^2)=\int_0^4 f(t)\,dt-0$$

$$\Rightarrow \int_0^4 f(t)\,dt=2\alpha f(\alpha^2)+2\beta f(\beta^2)$$

Hence proved.

Q23 **Prove that in a triangle, maximum value of $\sum \sin A=\frac{3\sqrt{3}}{2}$**

Sol. Let $y=\sin x, x\in(0,\pi)$

$\Rightarrow y''=-\sin x<0$

$\Rightarrow$ graph will be concave downwards.

Let there be 3 points $P(A,\sin A), Q(B,\sin B), R(C,\sin C)$

Centroid of the ΔPQR will be $G\left(\frac{A+B+C}{3},\frac{\sum \sin A}{3}\right)$

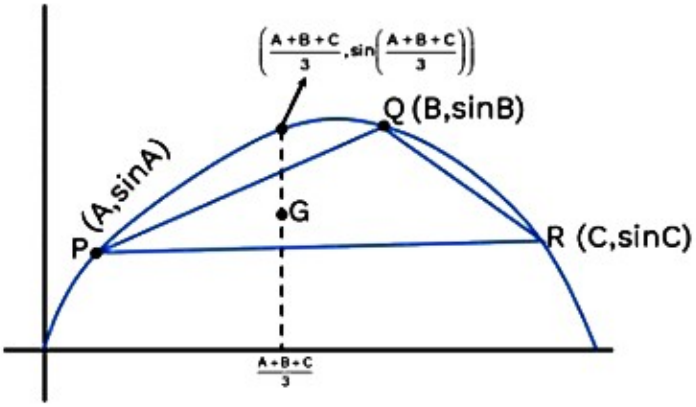
Since centroid always lie in the interior of a triangle, from graph we can say that

$$\frac{\sum \sin A}{3}<\sin\left(\frac{A+B+C}{3}\right)$$

$$\Rightarrow \sum \sin A<3\sin 60°$$

$$\Rightarrow \sum \sin A<\frac{3\sqrt{3}}{2}$$

But if A=B=C,
then all points coincides



$$\Rightarrow \sum \sin A = \frac{3\sqrt{3}}{2}$$

$$\Rightarrow \text{ Maximum value of } \sum \sin A = \frac{3\sqrt{3}}{2}.$$

Q24 **If $f(x)$ and $g(x)$ are continuous on [a, b] and derivable in (a, b) then show that**

$$\begin{vmatrix} f(a) & f(b) \\ g(a) & g(b) \end{vmatrix} = (b-a)\begin{vmatrix} f(a) & f'(c) \\ g(a) & g'(c) \end{vmatrix} \text{ where } a < c < b.$$

Sol. Let $h(x) = \begin{vmatrix} f(a) & f(x) \\ g(a) & g(x) \end{vmatrix}$

Applying LMVT on h(x) in [a, b],

$$h'(c) = \frac{h(b) - h(a)}{b-a} \text{ for some } c \in (a,b)$$

$$\Rightarrow (b-a)\begin{vmatrix} f(a) & f'(c) \\ g(a) & g'(c) \end{vmatrix} = \begin{vmatrix} f(a) & f(b) \\ g(a) & g(b) \end{vmatrix} - 0$$

Hence proved.

Q25 **$f'''(x)$ exists $\forall x \in R$ and $f(x) = f(6-x)$, $f'(0) = 0 = f'(2) = f'(5)$.**

Find minimum number of roots of equation $(f''(x))^2 + f'(x)f'''(x) = 0$ in [0, 6]

Sol. $f(x) = f(6-x)$

$x \to x+3$

$\Rightarrow f(3+x) = f(3-x)$

Differentiating, $f'(3+x) = -f'(3-x)$

Put $x = 0, \Rightarrow f'(3) = -f'(3)$

$\Rightarrow f'(3) = 0$

Also, $f'(0) = 0 \Rightarrow f'(3+(-3)) = -f'(3-(-3))$

$\Rightarrow f'(6) = 0$

Similarly, $f'(2) = 0 \Rightarrow f'(4) = 0$

$f'(5) = 0 \Rightarrow f'(1) = 0$

So, $f'(x) = 0$ for $x = 0, 1, 2, 3, 4, 5, 6$.

Since $f'(x) = 0$ has 7 roots, $f''(x) = 0$ will have atleast 6 roots in [0, 6]

(using Rolle's theorem)

So, $g(x) = f'(x)f''(x) = 0$ will have atleast 6 + 7 = 13 roots in [0, 6]

Again using Rolle's theorem on g(x),

$g'(x) = (f''(x))^2 + f'(x)f'''(x) = 0$ will have atleast

12 roots in [0, 6].

SOLVED EXAMPLES

Q1 **Find the values of a, if the equation $x - \sin x = a$ has a unique root in $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$.**

Sol. Consider the function

$f(x) = x - \sin x - a,\ x \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

Then $f'(x) = 1 - \cos x = 2\sin^2\left(\frac{x}{2}\right) \geq 0\ \forall\ x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

$\Rightarrow$ f(x) increases in $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

Also, we have $f\left(\frac{-\pi}{2}\right) = \frac{-\pi}{2} + 1 - a$ and $f\left(\frac{\pi}{2}\right) = \frac{\pi}{2} - 1 - a$

The curve y = f(x) will cut the X-axis exactly once, if $f\left(\frac{-\pi}{2}\right)$ is negative or zero

and $f\left(\frac{\pi}{2}\right)$ is positive or zero.

i.e. $\frac{-\pi}{2} + 1 - a \leq 0$ and $\frac{\pi}{2} - 1 - a \geq 0$

i.e. $a \geq \frac{-\pi}{2} + 1$ and $a \leq \frac{\pi}{2} - 1$

Hence, we have $a \in \left[1 - \frac{\pi}{2}, \frac{\pi}{2} - 1\right]$

Q2 **Find the intervals in which the given function increases or decreases**

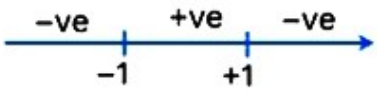
$f(x) = \dfrac{x^2 + x + 1}{x^2 - x + 1}$.

Sol. We have $f(x) = \dfrac{x^2 + x + 1}{x^2 - x + 1}, x \in R$

and $f'(x) = \dfrac{(x^2 - x + 1)(2x + 1) - (x^2 + x + 1)(2x - 1)}{(x^2 - x + 1)^2}$

$= \dfrac{(2x^3 - x^2 + x + 1) - (2x^3 + x^2 + x - 1)}{(x^2 - x + 1)^2} = \dfrac{-2(x + 1)(x - 1)}{(x^2 - x + 1)^2}$

Now, from the sign scheme for f'(x), we have



$\Rightarrow$ f(x) strictly decreases in $(-\infty, -1)$
strictly increases in $(-1, 1)$
strictly decreases in $(1, \infty)$.

Q3 **Prove that $1 + \cot x \le \cot \dfrac{x}{2} \forall x \in (0, \pi)$.**

Sol. Consider the function $f(x) = \cot\left(\dfrac{x}{2}\right) - 1 - \cot x, x \in (0, \pi)$.

Then $f'(x) = \dfrac{-1}{2}\operatorname{cosec}^2\left(\dfrac{x}{2}\right) + \operatorname{cosec}^2 x = \dfrac{1}{\sin^2 x} - \dfrac{1}{2\sin^2(x/2)}$

$= \dfrac{1}{2\sin^2(x/2)}\left[\dfrac{1}{2\cos^2(x/2)} - 1\right] = \dfrac{-\cos x}{4\sin^2(x/2)\cos^2(x/2)}$

$\Rightarrow f'(x) = \dfrac{-\cos x}{\sin^2 x}$, $f'(x) < 0 \ \forall x \in \left(0, \dfrac{\pi}{2}\right)$ $f'(x) > 0 \forall x \in \left(\dfrac{\pi}{2}, \pi\right)$

$\Rightarrow$ f(x) strictly decreases in $\left(0, \dfrac{\pi}{2}\right)$ and strictly increases in $\left(\dfrac{\pi}{2}, \pi\right)$

$\Rightarrow$ f(x) has least value at $x = \dfrac{\pi}{2}$

$\Rightarrow f(x) \ge f\left(\dfrac{\pi}{2}\right) = 0$

i.e. $\cot\left(\dfrac{x}{2}\right) \ge 1 + \cot x$. which proves the desired result.

Q4 **Let $f(x) = x^3 + ax^2 + bx + 5\sin^2 x$ be an increasing function on the set R. Then find the condition on a and b.**

Sol. $f(x) = x^3 + ax^2 + bx + 5\sin^2 x$ is increasing on R

$\Rightarrow$ $f'(x) > 0$ for $x \in R$

$\Rightarrow$ $3x^2 + 2ax + b + 5\sin 2x > 0$ for all $x \in R$

$\Rightarrow$ $3x^2 + 2ax + (b - 5) > 0$ for all $x \in R$

$\Rightarrow$ $(2a)^2 - 4 \times 3 \times (b - 5) < 0$

$\Rightarrow$ $a^2 - 3b + 15 < 0$

Q5 **Find the values of x where $f(x) = \sin(\ell nx) - \cos(\ell nx)$ is strictly increasing.**

Sol. Since $f(x) = \sin(\ell nx) - \cos(\ell nx)$, $x > 0$

$$= \sqrt{2}\sin\left(\ell nx - \frac{\pi}{4}\right)$$

$$\therefore \quad f'(x) = \frac{\sqrt{2}}{x}\cos\left(\ell nx - \frac{\pi}{4}\right)$$

$$= \frac{\sqrt{2}}{x}\sin\left(\frac{\pi}{2} + \ell nx - \frac{\pi}{4}\right) = \frac{\sqrt{2}}{x}\sin\left(\frac{\pi}{4} + \ell nx\right) > 0 \qquad (\because x > 0)$$

or $\sin\left(\frac{\pi}{4} + \ell nx\right) > 0$ or $2n\pi < \frac{\pi}{4} + \ell nx < (2n+1)\pi, n \in I$

or $2n\pi - \frac{\pi}{4} < \ell nx < 2n\pi + \frac{3\pi}{4}, n \in I$ $\Rightarrow$ $e^{2n\pi - \frac{\pi}{4}} < x < e^{2n\pi + \frac{3\pi}{4}}$, $n \in I$

Therefore, f(x) is strictly increasing when $x \in \left(e^{2n\pi - \frac{\pi}{4}}, e^{2n\pi + \frac{3\pi}{4}}\right), n \in I$

Q6 **Prove that $\sin^2\theta < \theta\sin(\sin\theta)$ for $0 < \theta < \frac{\pi}{2}$.**

Sol. In this problem, first we have to select an appropriate function.

Now by observation, given inequality can be set as $\frac{\sin(\sin\theta)}{\sin\theta} > \frac{\sin\theta}{\theta}$.

This clearly gives indication that one has to study the function $f(x) = \frac{\sin x}{x}$

$\Rightarrow f'(x) = \frac{(x\cos x - \sin x)}{x^2} = \frac{\cos x(x - \tan x)}{x^2} < 0$ (as in first quadrant $x < \tan x$)

$\Rightarrow$ f(x) is a decreasing function

Now, $\sin\theta < \theta$ for $0 < \theta < \frac{\pi}{2}$

$\Rightarrow f(\sin\theta) > f(\theta) \Rightarrow \frac{\sin(\sin\theta)}{\sin\theta} > \frac{\sin\theta}{\theta}$ [From (i)]

Hence, $\sin^2\theta < \theta \sin(\sin\theta)$ for $0 < \theta < \frac{\pi}{2}$.

Q7 **Let f : R → R be a function such that f(x) = ax + 3 sin x + 4 cosx. Then f(x) is invertible if**

(A) a ∈ (−5, 5) **(B) a ∈ (−∞, 5)** **(C) a ∈ (−5, + ∞)** **(D) None of these**

Sol. **(D)**

$f'(x) = a + 3\cos x - 4\sin x = a + 5\cos(x + \alpha)$, where $\cos\alpha = \frac{3}{5}$

For invertible, f(x) must be monotonic

$\Rightarrow$ $f'(x) \geq 0\ \forall\ x$ or $f'(x) \leq 0\ \forall\ x$

$\Rightarrow$ $a + 5\cos(x + \alpha) \geq 0$ or $a + 5\cos(x + \alpha) \leq 0$

$\Rightarrow$ $a \geq -5\cos(x + \alpha)$ or $a \leq -5\cos(x + \alpha)$

$\Rightarrow$ $a \geq 5$ or $a \leq -5$

Q8 **f(x) = (x − 2) |x − 3| is monotonically increasing in**

(A) $\left(-\infty, \frac{5}{2}\right) \cup (3, \infty)$ **(B)** $\left(\frac{5}{2}, \infty\right)$ **(C) (2, ∞)** **(D) (−∞, 3)**

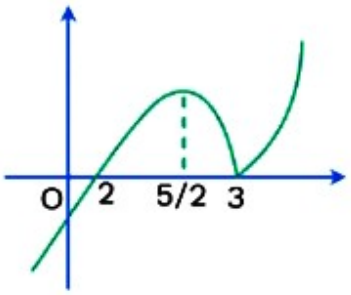
Sol. **(A)**

$f(x) = (x - 2)|x - 3|$

For $x \geq 3$,

$f(x) = (x - 2)(x - 3) = x^2 - 5x + 6$

$f'(x) = 2x - 5 = 0 \Rightarrow x = \frac{5}{2}$ (Not possible as $x \geq 0$)

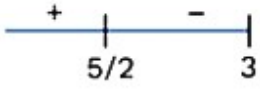


For $x < 3$ $f(x) = -(x - 2)(x - 3)$

$f'(x) = 0 \Rightarrow x = 5/2$

Now, the graph of $f(x) = (x - 2)|x - 3|$ is drawn

Clearly from the graph, f(x) increases in $\left(-\infty, \frac{5}{2}\right) \cup (3, \infty)$.



Q9 **A function g(x) is defined as $g(x) = \frac{1}{4}f(2x^2-1) + \frac{1}{2}f(1-x^2)$ and f'(x) is an increasing function, then g(x) is increasing in the interval**

(A) (–1, 1)

(B) $\left(-\sqrt{\frac{2}{3}}, 0\right) \cup \left(\sqrt{\frac{2}{3}}, \infty\right)$

(C) $\left(-\sqrt{\frac{2}{3}}, \sqrt{\frac{2}{3}}\right)$

(D) None of these

Sol. **(B)**

$g'(x) = xf'(2x^2-1) - xf'(1-x^2) = x(f'(2x^2-1) - f'(1-x^2))$

$\Rightarrow g'(x) > 0$

(i) if $x > 0$, $2x^2 - 1 > 1 - x^2$ (as f' is an increasing function)

$$\Rightarrow 3x^2 > 2 \Rightarrow x \in \left(-\infty, -\sqrt{\frac{2}{3}}\right) \cup \left(\sqrt{\frac{2}{3}}, \infty\right) \Rightarrow x \in \left(\sqrt{\frac{2}{3}}, \infty\right)$$

(ii) If $x < 0$, $2x^2 - 1 < 1 - x^2$

$$\Rightarrow 3x^2 < 2 \Rightarrow x \in \left(-\sqrt{\frac{2}{3}}, \sqrt{\frac{2}{3}}\right) \Rightarrow x \in \left(-\sqrt{\frac{2}{3}}, 0\right).$$

$$\therefore x \in \left(-\sqrt{\frac{2}{3}}, 0\right) \cup \left(\sqrt{\frac{2}{3}}, \infty\right)$$

Q10 **Let $f(x) = 1 - x - x^3$. Find all real values of x satisfying the inequality, $1 - f(x) - f^3(x) > f(1 - 5x)$**

Sol. $f(x) = 1 - x - x^3 \quad \Rightarrow f'(x) = -1 - 3x^2$ which is –ve $\forall x \in R \Rightarrow$ f is decreasing

$f[f(x)] = 1 - f(x) - f^3(x)$

$\therefore \quad f[f(x)] > f(1 - 5x)]$ given

since, f(x) is decreasing hence

$f(x_1) > f(x_2) \Rightarrow x_1 < x_2$

$\therefore \quad f(x) < 1 - 5x$

$1 - x - x^3 < 1 - 5x$

$x^3 - 4x > 0$

$x(x^2 - 4) > 0$

$\therefore \quad x \in (-2, 0) \cup (2, \infty)$

